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# ASSET INTEGRITY TECHNICAL DOCUMENTATION

**PROJECT:**

## SBM OFFSHORE

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|---|---|---|---|---|---|

**DOCUMENT DESCRIPTION:**

# MANAGING MAINTENANCE OPTIMIZATION PROGRAM

# E3 PROJECT

# BUSINESS RULES

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|---|---|---|---|---|---|
| A1 | 05-01-24 | 34 | **Hugo Chiarelo** Reliability Engineer | **Agatha Alvarenga** Reliability Lead | **Cid Costa** Reliability Manager |
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**FOR INFORMATION:**

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|---|---|
| **P** | **Preliminary Issue** |
| **I** | **Inter Discipline Checking** |
| **C** | **For Comments and/or Approval** |
| **V** | **Valid for Construction** |
| **A** | **Approved** |
| **D** | **Deleted** |
| **X** | **"As Built"** |

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## REVISION STATUS / SUMMARY OF CHANGES

| REVISION | REVISED CHAPTERS | REVISION DESCRIPTION | REASON FOR REVISION |
|---|---|---|---|
| A1 | Fist emission | For issue | For Issue |
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PURPOSE
This document aims to describe concepts, criteria, and standards for E3 Project.

The main objective of E3 is to update maintenance plans to achieve optimized and standardized activities according with EMS based on 3 pillars:

- Eliminate – remove non-value preventive tasks
- Elevate – Check whether continuous monitoring already exists (ASI) and can be taken as credit for eliminating preventive tasks
- Enrich – Verify and update the attributes:
  - Item Class (incorrect or missing)
  - Category
  - Object Type and Criticality (if needed)
  - Process Code
  - Object creation (If it's impacting the strategy application)
  - PM Performed Date Based
  - PM Interval
  - PM Interval Unit
  - PM Leadtime
  - PM Action
  - PM Priority
  - PM Execution time (Standardized as WTT)
  - Rounds
  - PM Maintenance Organization
  - PM Operational Status
  - PM Resource
  - PM Status
  - PM Creation/ Status Review (If needed)

What is not part of Enrich:
- PM Third Party Required flag
- PM Shutdown Required flag
- PM WOs parameter updates (Priority, Latest Finish and Maintenance Org)

## 1. ABBREVIATIONS

- 3YP - Three Years Plan
- CMMS - Computerized Maintenance Management System
- EDMS - Electronic Document Management System (SBM documentation)
- FPSO - Floating Production Storage & Offloading
- IFS - Industrial and Financial Systems
- IWP - Integrated Work-Package
- KPI - Key Performance Indicator
- PM - Preventive Maintenance
- POB - People Onboard
- WTT - Work Task Template
- ASI – Asset Surveillance Intelligence

## 2. W4W – MAINTENANCE STRATEGY OPTIMIZATION PROGRAM

### 2.1 SCOPE OF WORK

The workflow below defines the steps to be carried out during E3 Program.

The program consists of guarantees:

- The right routine in right asset.
- The PM action attributes priority, interval and Leadtime.

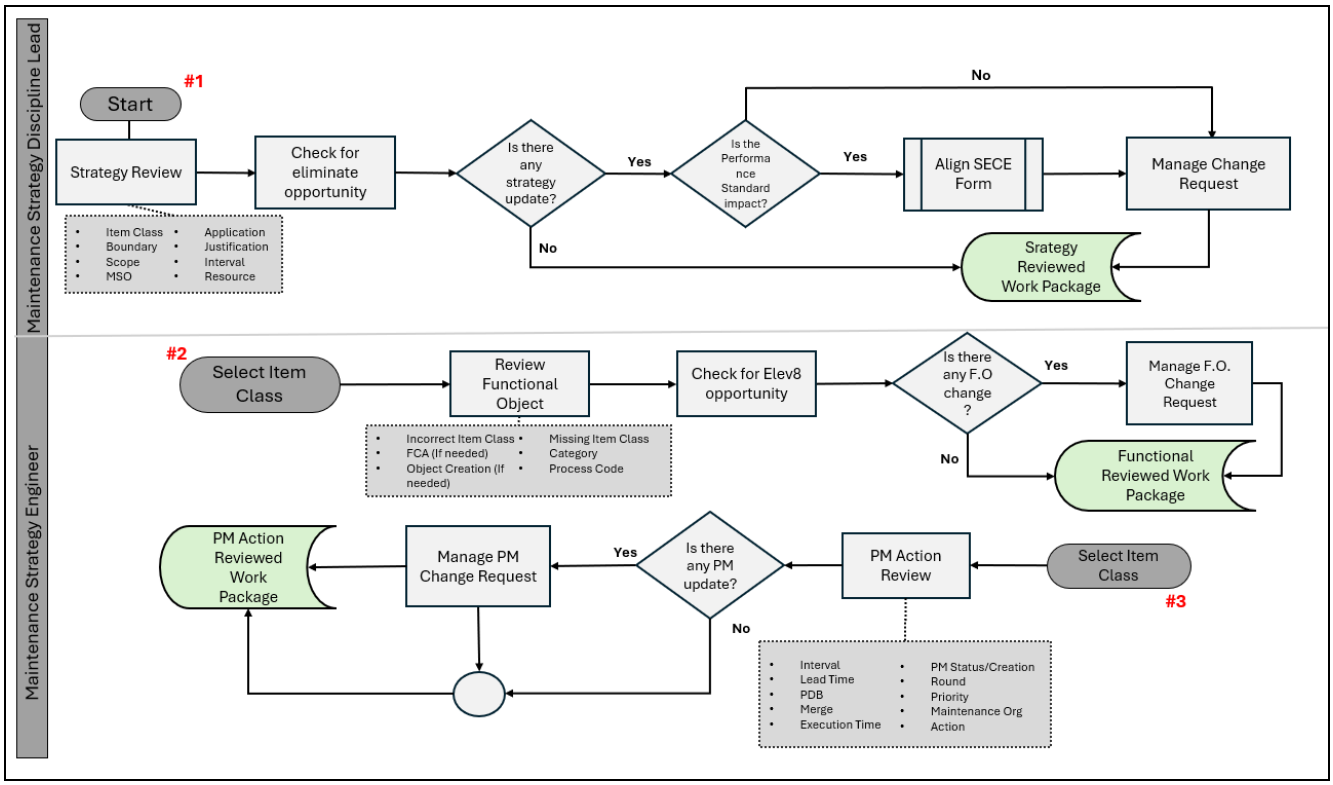



Figure 1 - E3 Workflow

The workflow above specifies the steps for analyzing the item class, which starts with the review of the Strategy, then Functional Object features and up to the adjustments in the PM Actions.

## 2.2 PREMISSES

### a) Eliminate

The project has a goal to eliminate tasks with no value.
For this reason, the items below should NOT be part of CMMS:

- Operational Tasks (Non-maintenance)
- Day-by-day routine
- Tasks that don't cover any failure mode
- Checklists or alarms checks
- Procedures
- Class, or flag certification
- Operational routines used 'reminder'

For new PM Actions created during the E3 project, this is the schedule recommendation:

- New PM actions created, if there's historical, should be scheduled according to the last completed date.

- If it's a whole new PM and P1, at least 60 days should be assumed in the scheduling or when aligned with onboard/planner team a feasible date not more than 60 days in order to may assume an ORA.

- For the new PM Actions, that are P2 or P3, to choose the first execution dates, feedback from the onboard/planner team should be considered and based on the WOs execution forecast for the following months (not more than 6 months).

## 2.3 ITEM CLASS

The Item Class is an attribute to describe the functionality of the equipment within the unit process as a unique 4-caracter sequence.

## 2.4 EMS

### 2.4.1 Boundary

Boundaries: associated Maintainable Item that can be included or excluded for Maintenance management purposes.

Definition of Item Class different types must be inside the Scope field.

Do not mention specific Item Class, use generic references. Where the specific Item class can be mentioned inside the “Other Associated Item Classes”

For better standardization of the "Boundary" field writing use “Include” or “Exclude”.
Parts/Maintainable Items that introduce different failure modes to the Item Class should not be on the same line of the Boundary.
Example:

*“Excludes Monitoring and control systems”*
*“Include Cooling system”*

### 2.4.2 Maintenance Strategy Overview

This field consolidates the criteria, premises, and technical decisions that guided the construction of the maintenance plan for each Item class. The strategy is based on a combination of standards, historical reliability and maintenance data, expected failure modes, required performance (e.g., availability, safety, regulatory compliance), and operational context.
The overview should address the following guidance:

- **Maintenance Basis:** Is maintenance based on time, condition, predictive analytics, or run-to-failure?

  To increment the Maintenance basis information, if applicable, use the following information:

  - **Logic for Task Selection and Frequency:** What methodology (e.g., RCM, RBI, OEM, regulatory, performance standard) was used to define the types and frequencies of tasks?
  - **Operational Assumptions:** What are the key operational or environmental assumptions that influence strategy (e.g., duty cycle, standby, redundancy, environmental conditions)?
  - **Integrity and Assurance Programs:** Is the equipment part of an integrity management or assurance program (e.g., safety case, SIL, performance standard)?

These preventive inspection and maintenance tasks will then be applied at different frequencies, depending on the potential criticality and risk profile of the equipment, forming the basis of the generic Risk-Based Maintenance approach defined by SBM.
For package strategies that cover equipment with their own Item Class, the strategies for that equipment should be mentioned generically within the package's Maintenance Strategy Overview, without specific mention of Item Class or Work Task Template. Example:

*“Diesel engines have their own strategy.”*

Specific Item Classes will be mentioned within "Other Associated Item Class".

#### 2.4.3 Scope

Define which functions the equipment is supposed to work (it might include the function of the auxiliary equipment if relevant).

#### 2.4.4 Other Associated Item Class

Described using Item Class codes based on the Boundary
Example:

DRDE – Diesel Engine

#### 2.4.5 Define Work Task Template

A Work Task Template is a standardized model that describes How, When, Where, Why and by Whom a maintenance activity should be performed.

#### 2.4.6 Work task template Justification

The Justification field defines the rationale for each maintenance activity and its frequency. It ensures traceability to the underlying driver for the task, supporting compliance, risk management, reliability, and performance objectives. The justification should reference one or more of the following guidances:

- **Regulatory Requirement:** Mention the specific regulation, standard, or legal requirement (including name, section, paragraph) mandating the activity.
- **Assurance Activity / Performance Standard:** Reference the performance standard or safety case requirement and explain how the activity assures compliance.
- **Maintenance Alert Outcome:** Reference any maintenance alert, bulletin, or safety notice that drives the activity.
- **Cost effective:** This maintenance activity is cost-effective by reducing unplanned failures, downtime, and overall maintenance costs.
    - Industry Recommended Practice: Reference the issuing body and relevant guideline or best practice.
    - Reliability/Risk Driven: Summarize the outcome of reliability studies (RCM, RBI, FCA, SIL, etc.), risk assessments, or criticality analyses that justify the activity.
    - Failure History/Experience: Reference historical failure data, bad actor analysis, or incident investigation that led to the activity.
    - OEM Driven: Reference the OEM, relevant documentation, and summarize the technical rationale or outcome.

#### 2.4.7 Application

The Application field defines the scope and boundaries for where and when a specific maintenance job should be applied. It ensures that maintenance activities are targeted to the correct equipment, locations, and operational contexts, and that any exclusions or special conditions are clearly communicated. The Application field should follow the guidance:

- **Scope of Applicability:**
    - Fleet-wide, region-specific, unit-specific, model-specific, design-specific, function-specific, or plan-specific.

- **Conditional Applicability:**
  - Any operational, environmental, or regulatory conditions that must be met for the job to apply (e.g., “only when equipment is in continuous service,” “applicable only to units with external turret”).

- **Exclusions/Exceptions:**
  - Where the job should NOT be applied (e.g., “not applicable to AMETEK analysers,” “exclude submersible pumps”).

- **Activation/Deactivation Guidance:**
  - Specify if and when the job should be applied (e.g., “deactivate if managed by 3rd party,” “do not apply if included in package routine”).

- **Reference to Source:**
  - If applicability is driven by a standard, OEM, or risk assessment, reference the source document or study.

- **Instructions for Planning:**
  - Only if the WTT is eligible for Replacement functionality. Standard phrase to indicate the Replacement:

*“Eligible for Replacement Functionality”*

This is a proposal standard for application implementation in the Wok Task Template:

*“Applicable [scope of applicability], in accordance with [procedure/standard/reference].*
*[Special conditions/Equipment Model or attention points].*
*[Instructions for planning, activation, or deactivation]”.*

### 2.4.8 Interval

The Maintenance Interval for each equipment is defined at the Work Task Template (WTT) level. The interval is set in different instances depending on the criticality of the Object to which the WTT will be applied. The criticality levels of an Object are defined as L (Low), M (Medium), and H (High), and each may have its own maintenance interval, which may be the same or different between them. When the interval is left blank for all criticality levels, it means the interval is interpretative according to a standard, norm, or internal SBM definition—usually described in the “Justification” field.

If at least one of the interval fields is filled in, it means that objects with blank criticality fields should not receive that specific WTT.

For situations where the Work Task Template has different intervals for the same equipment criticality, the intervals and their justifications must be described in the “Justification” field, and the Interval field should be left blank.

Functions | Child Item Classes | EMS

| WTT ID | WTT status | WTT Definition | Action Description | Assurance Activity | 3rd Party Required | High Risk Frequency | Unit High | Medium Risk Frequency | Unit Medium | Low Risk Frequency | Unit Low |
|---|---|---|---|---|---|---|---|---|---|---|---|
| EXXON028A | Active | 1Y LINE FLUSHING | Test | FALSE | FALSE | 12 | Month | 12 | Month | | |
| EXXON027A | Active | 1Y VISUAL INSPECTIONv PT | Inspection | FALSE | FALSE | 12 | Month | 12 | Month | | |
| EXXON029A | Active | 2Y VISUAL INSPECTION | Inspection | FALSE | FALSE | 24 | Month | 24 | Month | | |
| PM00305B | Active | Pressure Transmitter Risk Based Fun Test | Test | TRUE | FALSE | | | | | | |

Figure 2 - EMS interval

When the criticality level is indicated as "Low" (empty), it means that this strategy does not apply to this type of low-critical equipment, and it should be treated as a "Run to Failure" category device.

## 2.5 IMPLEMENTATION

The E3 implementation consist in a tool aims to assist Maintenance Strategy engineers in the activities of reviewing and standardizing the maintenance strategy in the CMMS.

The tool has automations to ensure and facilitate the replicability of the maintenance strategy, as well as generate history and calculate the compliance rate by Item Class, giving visibility to the gaps found.

Below, the flow of activities for using the tool is shown.

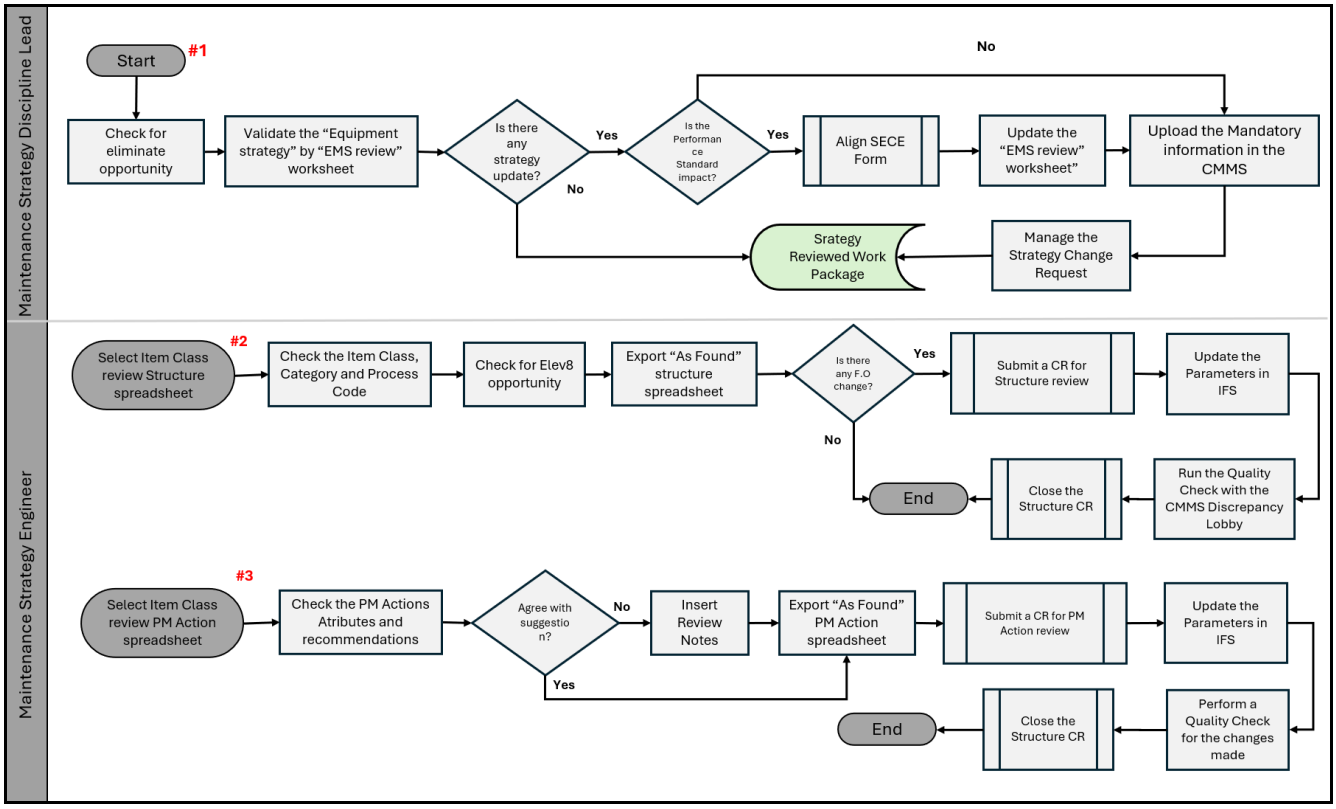



*Figure 3 - Detailed Workflow of W4W Process*

To export spreadsheets from IFS, there are some standard Queries, this Queries extraction will be under "Project Manager responsibility".

The Structure and PM Action review spreadsheets are available for each discipline inside the Month analysis folder.

In Figures 4 & 5 there are the customized reports built within the "Query Builder" module.

Below is the Query model built to export information from Functional Objects, the first part of analysis indicated in the flow.

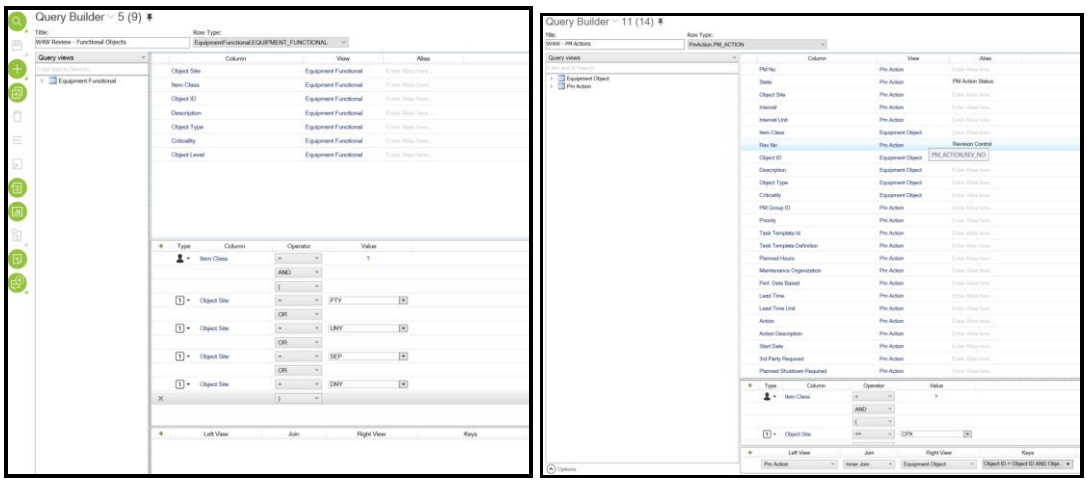
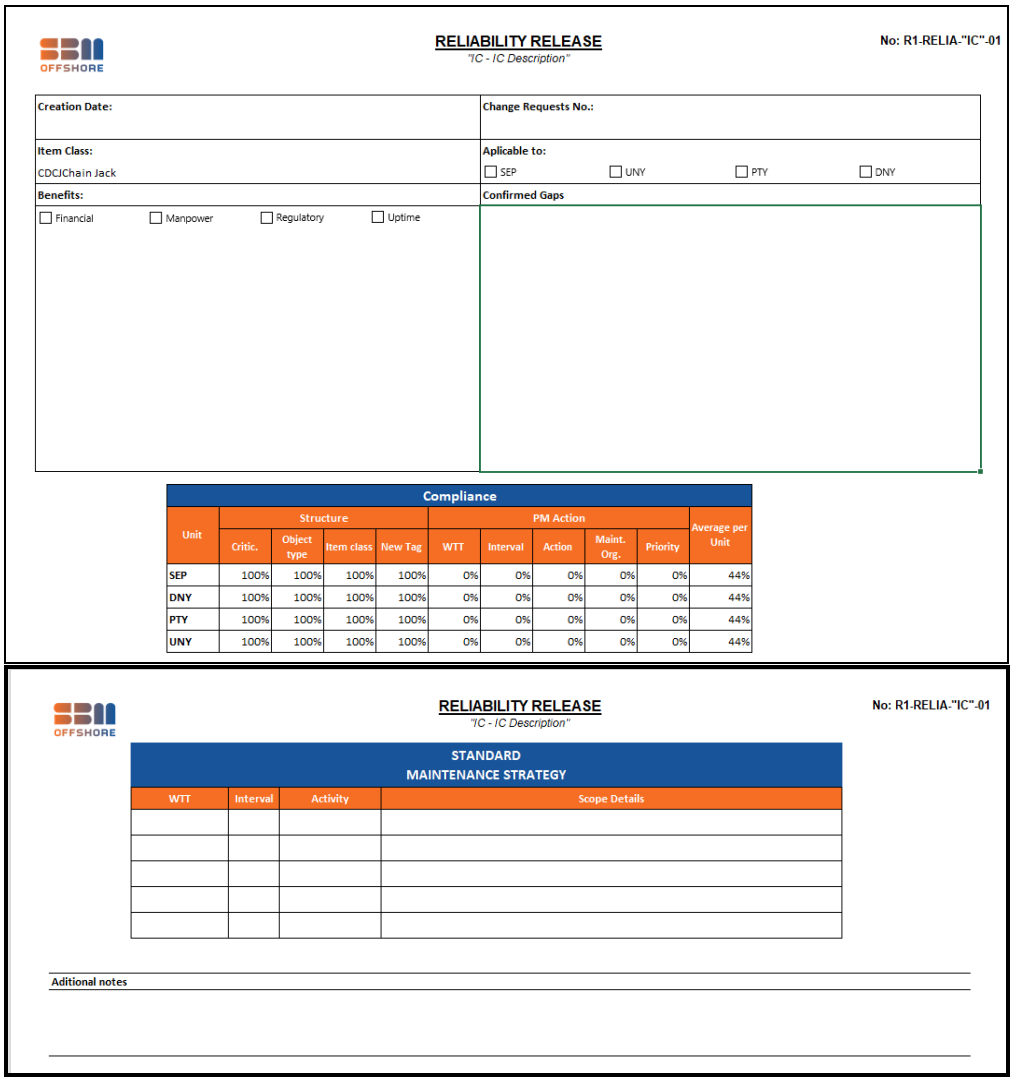


*Figure 4 & 5 – Example of Report in the Query Tool for Functional Objects & PM Actions*

During the workflow, to ensure that there is traceability of the history and survey of compliance by item class, there are some steps in which the "as found" and "as left" extraction is performed.

This tool's will also give the output of “Reliability Release”, a document that highlights the main benefits of the analysis, the motivations, where it is applicable, the main gaps found in the analysis, the compliance index (As Found x As Left) and a summarized view of the strategy adopted for that Item Class. See the model used below:



**RELIABILITY RELEASE**
*"IC - IC Description"*

No: R1-RELIA-"IC"-01

| | |
|---|---|
| **Creation Date:** | **Change Requests No.:** |
| **Item Class:**<br>CDCJChain Jack | **Aplicable to:**<br>☐ SEP ☐ UNY ☐ PTY ☐ DNY |
| **Benefits:**<br>☐ Financial ☐ Manpower ☐ Regulatory ☐ Uptime | **Confirmed Gaps** |

| Compliance | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit | Structure | | | | PM Action | | | | | Average per Unit |
| | Critic. | Object type | Item class | New Tag | WTT | Interval | Action | Maint. Org. | Priority | |
| SEP | 100% | 100% | 100% | 100% | 0% | 0% | 0% | 0% | 0% | 44% |
| DNY | 100% | 100% | 100% | 100% | 0% | 0% | 0% | 0% | 0% | 44% |
| PTY | 100% | 100% | 100% | 100% | 0% | 0% | 0% | 0% | 0% | 44% |
| UNY | 100% | 100% | 100% | 100% | 0% | 0% | 0% | 0% | 0% | 44% |

**RELIABILITY RELEASE**
*"IC - IC Description"*

No: R1-RELIA-"IC"-01

| STANDARD MAINTENANCE STRATEGY | | | |
|---|---|---|---|
| WTT | Interval | Activity | Scope Details |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

Aditional notes

*Figure 6 – Reliability Release*

The Compliance table is automatically filled in based on the engineer's analysis carried out throughout the process, while the other fields are free to fill in.

During the initial phase of the project, certain parameters from PM Actions can be managed in bulk through a migration file, as they are either defined at the Work Task Template level or follow a clear and standardized application pattern. These are the parameters:

1. PM Operational Status
2. PM “Action”
3. PM Lead Time
4. PM Execution Time
5. PM Maint Organization

These parameters will subsequently undergo systematic reassessment throughout the project lifecycle as part of established quality assurance and compliance checks.

## 2.6 QUALITY CHECK

The Quality Check process ensures that all maintenance strategies are implemented in the CMMS comply with the business rules, standards, and project requirements defined during the planning phase. This step is critical to guarantee consistency, regulatory compliance, and reliability across the fleet.

The result of the Quality Check is a compliance report that measures adherence to project rules and identifies any deviations. Discrepancies are documented and addressed through corrective actions to the responsible discipline engineer. This ensures that the CMMS reflects an accurate, standardized, and auditable maintenance program.

## 2.7 FUNCTIONAL OBJECT ATTRIBUTES

### 2.7.1 Criticality / Object Type

For criticality (Object Type and Criticality) assessment, the FCA procedure should be followed [4]. But any change should be linked and aligned in Change Request if it's out of the project.

When a change is required in the Object Type or Criticality, it's important to update the tab "Functional Criticality Assessment" accordingly the "*Assess Functional Criticality*" procedure, the values to be updated are the Likelihood, Severity, Level… See an example below:

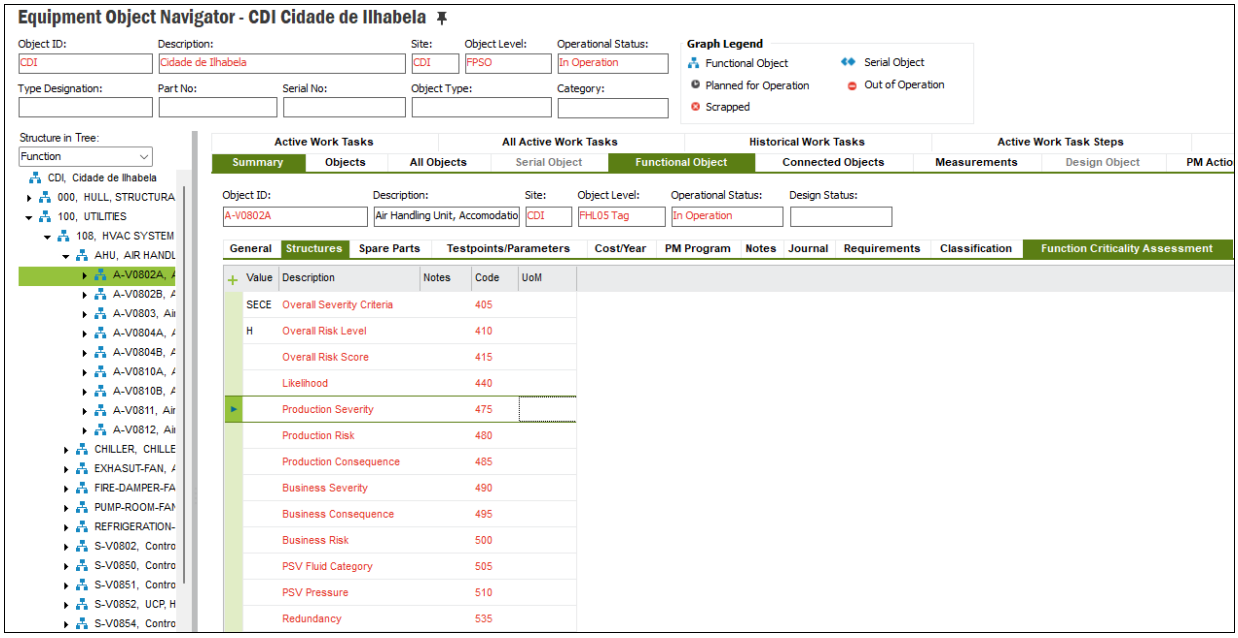



*Figure 7 - FCA Tab for Functional Object*

### 2.7.2 Operational Status

The operational status of the object in its lifecycle. Possible values used in the CMMS are:

- **Planned for Operation**: The object has been transferred from IFS/Project Delivery or IFS/Asset Design and has not yet been set to In Operation.
- **In Operation**: The object is in use.
- **Out of Operation**: Equipment out of operation due to a BM Work Order – Loss of function and Under Long Term Isolation (LTI)
  ***Note: LTI during maintenance execution should not be considered for Equipment Status change.*
- **Scrapped**: Equipment Decommissioned, and no date scheduled to return or permanently out of use. Due to management decisions, or lack of need.

**Note: *Metering equipment with temporary/permanent unavailability should be considered Scrapped.*

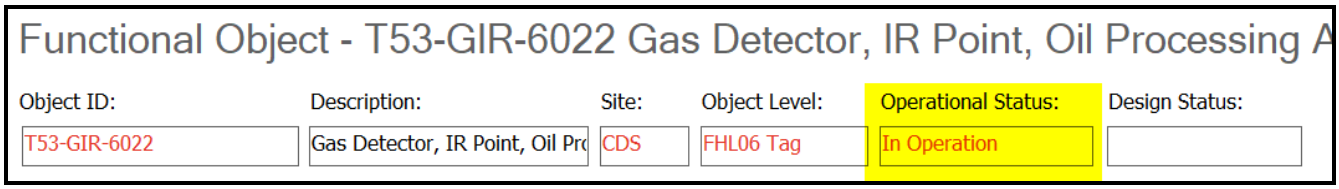



*Figure 8 – Object Operational Status*

For case where the Object is moved to "Scrapped", the change request should be opened by the responsible onboard and approve the first step, in order to guarantee the responsibility of the information.

The items that need to be changed to "Scrapped" cannot remain with the status "Out of Operation."
These cases should only be adjusted when it is possible to migrate them to "Scrapped," and coordinating this action with the onboard team using project outside Change Request is necessary.

The Reliability Engineer will change to scrapped and guarantee that the Object is connected to the Change Request. For the Project, the Change Request used for this, is the one related to the Stratucture Analysis – *"W4W Item Class Analysis".*

### 2.7.3 Process Code

The Process Code is used to indicate which Performance Standard the object is classified as a safety barrier. So, it's a Functional Object element, that can be found in the
The Process Code is mandatory for all objects classified with the Object Type "SECE".
To create a Process Code, it is necessary to open a Change Request of the type: *"New/Update Equipment".*

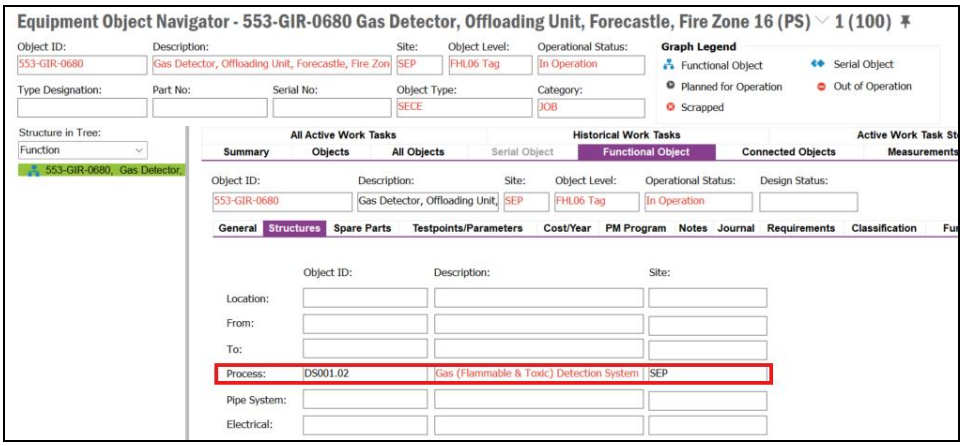



*Figure 9 – Process Code position*

### 2.7.4 Category

Category field is a Functional Object attribute that defines what kind of maintenance this function receives. The instruction for this categorization comes from EMS Guidance.

The Category can be:

| Code | Code Description | Definition |
|---|---|---|
| MHL | Maintenance High Level | MHL is a strategy applied when a component or sub-equipment is maintained indirectly, through its high-level parent equipment's maintenance plan. |
| MLL | Maintenance Low Level | MLL is a strategy applied when a component or sub-equipment is maintained indirectly, through its low-level parent equipment's maintenance plan. |

| | | |
|---|---|---|
| RTF | Run to Failure | Run to Failure (RTF) is a maintenance strategy where an asset is allowed to operate until it breaks down, with no preventive maintenance performed in advance. Usually applied for equipment with low criticality. |
| JOB | Job Assigned | JOB means that this Functional Object receives the PM actions according with EMS guidance. |
| NSM | Non Schedule Maintenance | NSM (Non-Scheduled Maintenance) is a classification used to indicate that no preventive maintenance is required or planned for a particular equipment or component.<br><br>It means the equipment is:<br>-Monitored passively (e.g., through operator observation and operational checklist – to be align into the EMS)<br>-Or the task is planned according to a specific opportunity or situation. For example, when there is work with a diver, the diving station needs to be certified. |
| STA | Monitored by Surveillance | MRU (Monitoring Rules) refers to a category designated for situations where at least one specific surveillance activity is implemented. This category ensures that Preventive Maintenance (PM) Actions are not excluded but can be integrated as needed to maintain operational safety and performance. |
| RBM | Risk Based Maintenance | RBM (Risk Based Maintenance) is a category used when any inspection or evaluation is performed under the framework of RBI, API 14C risk-based strategy, and/or SIL. This category includes assets with maintenance intervals justified by the consequence and likelihood of failure, rather than fixed schedules. |

*Table 1 – Category list.*

## 2.8 PM ACTION ATTRIBUTES

### 2.8.1 Performed date Based Standard

Displays whether the maintenance plan is to be adjusted in time, based on the actual completion date of the last connected work task on the work order that was generated from the PM action.
The completion date is based on when the last work task is set to status Work Done.

Notice that when you run a PM Action with Performed Date Based = 'Yes', only one Work Order will be allowed to be generated at a time.

The standard for the project is:

- Performed Date Based “Yes”: Use for any type of PM Action, except for weekly PMs.
- Performed Date Based “No”: Used only for PMs with weekly frequencies (7 days).

When choosing the PM Action in Performed Date Based "No" or "Yes" the only parameter to consider is frequency.

It is not necessary to consider whether there is Merge, Assurance Activity, or any other parameter, just the frequency.

It’s not necessary define inside the Work Task template Module the Performed Date Based, since the frequency may vary according to the criticality of the asset.

This attribute is properly defined at the PM Action level.

### 2.8.2 Changing Performed date Based from “No” to “Yes” Standard

During the review of PM Actions, one may encounter the situation where must perform the change from Performed Date Based (PDB) "No" to "Yes".
In the case of this specific change, the problem caused is when in the Maintenance Plan there is more than one open Work Order (Any status before Work Done).

This is a problem because in PDB "No", when there are more than one open Work Orders, will always have more than one maintenance line generating Work Orders varying according to the closing date, which can cause to have two equal tasks being carried out for the same equipment on the same date.

In these cases, the engineer must cancel the most recent ones and keep only the oldest one.

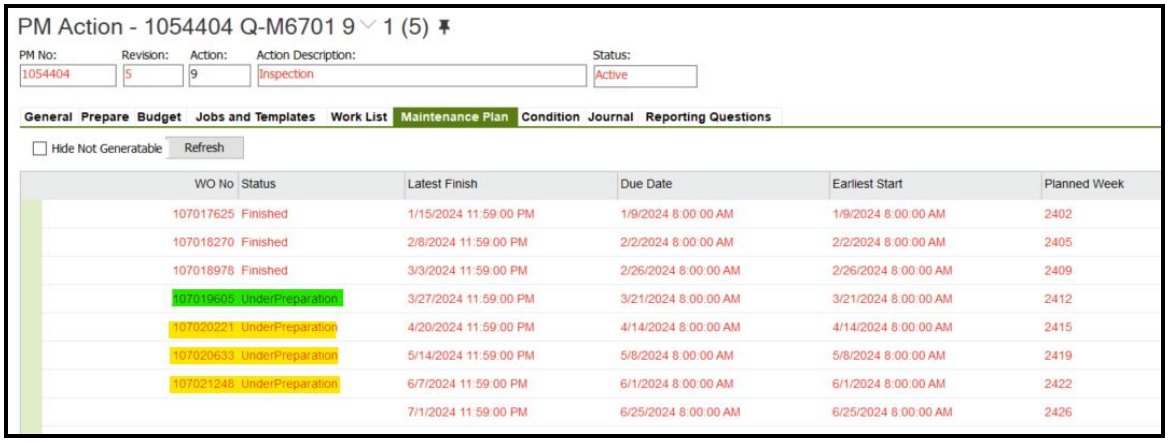



*Figure 10 – Changing from PDB "No" to "Yes"*

For the case above, the WOs in yellow must be canceled and only the one highlighted in green must be kept. To know how to cancel a WO, consult the Chapter "*2.9.6 – Cancelling of Work Order".*

For these when exists the need to cancel some Work Orders, it's mandatory to cancel using the "Cancel related work…" function – Due to keep the Maint Plan lines clean for the next WO generation.

There should be no concern regarding the date of the next WO or whether it will even be generated, since when the only remaining Work Order is done, the maintenance plan will follow the Last Work Task Completion Date and recalculate the plan.

### 2.8.3 Changing Performed date Based from "Yes" to "No" Standard

During the review of PM Actions, one may encounter the situation where must perform the change from Performed Date Based (PDB) "Yes" to "No".

In the case when there is an open WO, it's necessary to review the "Start Value" to match Maint Plan next available line. Because PDB "No" follows the start value defined in the PM Action.

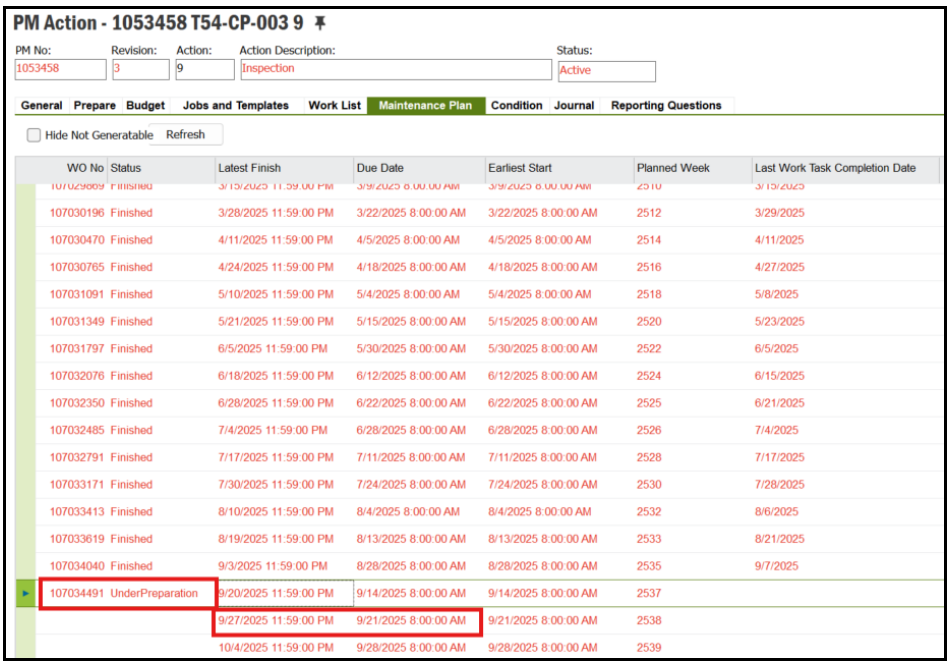



*Figure 11 – Changing from PDB "Yes" to "No"*

In this situation, the Open Work Order can be kept, but the start value needs to be aligned with the next maintenance plan line available.

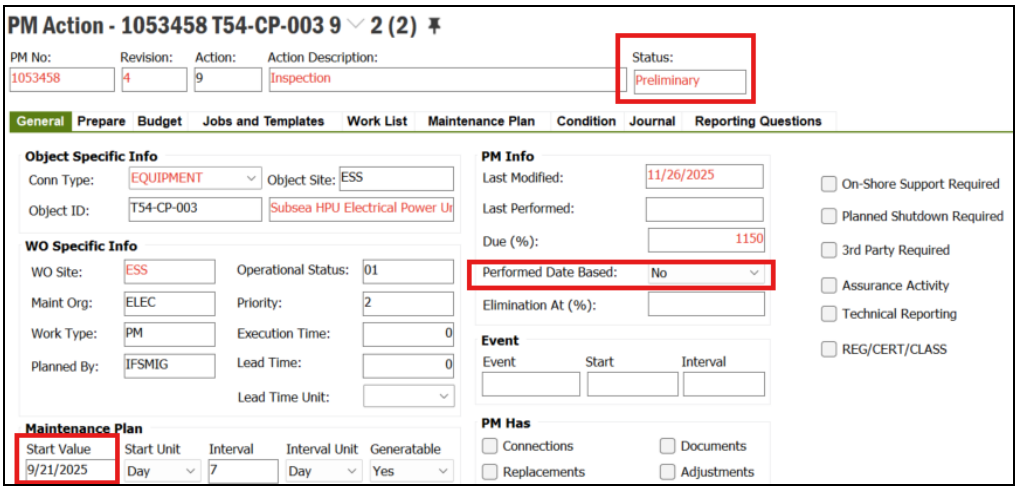



*Figure 12 – Changing from PDB "Yes" to "No" start value change*

The "Start Value" to be aligned with the next Main plan line should follow the "Due date" value.

### 2.8.4 Running Hours PM

Running Hours PMs are for equipment that receives Preventive Maintenance where the maintenance interval is based on hours and generated following the equipment's hour meter recorded in the equipment's own "Measurements" section.

Some settings are different for Running Hours PMs; the "General" tab of the PM should not be filled with an interval or Start Value. This information will be in the "Condition" tab.

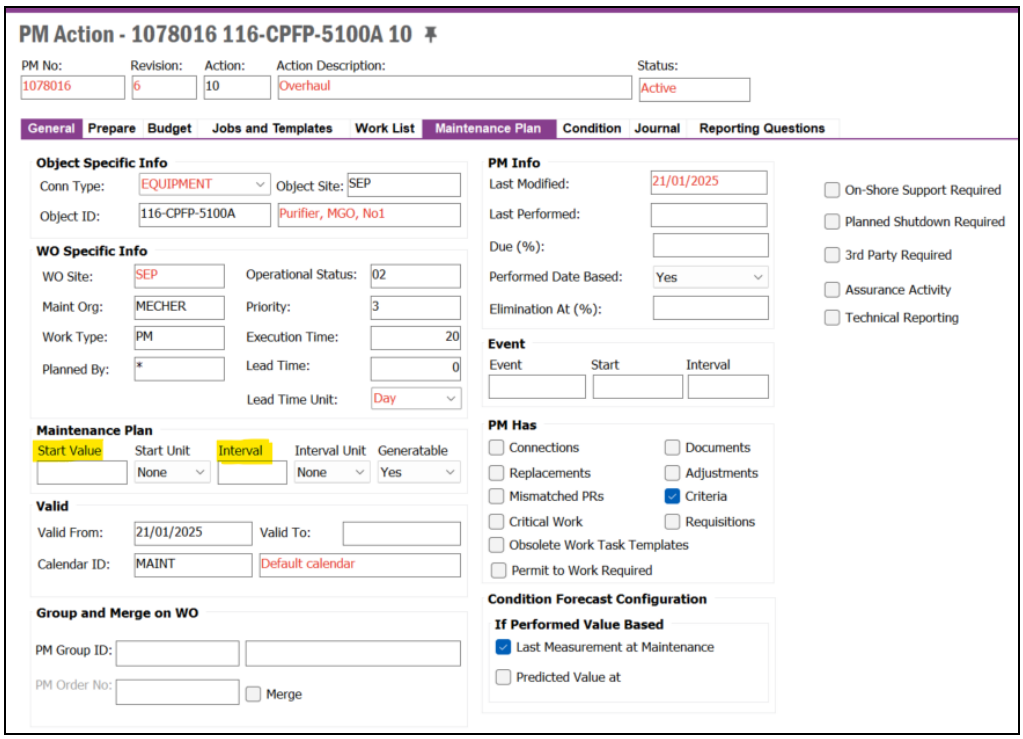



*Figure 13 – Running Hours Interval*

In the "Condition" tab, the following parameters must be filled in by the engineer:

- **Test Point ID:** Fill in with the default value "*" for recording in the objects.
- **Parameter ID:** Fill in with the default value "RH" to represent the equipment's hour meter.
- **Start Value:** Represents the generation value (hours) of the first maintenance cycle; when building the PM Action, this value should be that of the next WO generation.
- **Interval:** Follow the interval designated by the EMS (hours).
- **Performed Value Based:** Represents whether the WO generation will be variable or fixed - By default, always choose "Yes" that represents Variable date.

The Other parameters are filled automatically by the IFS.

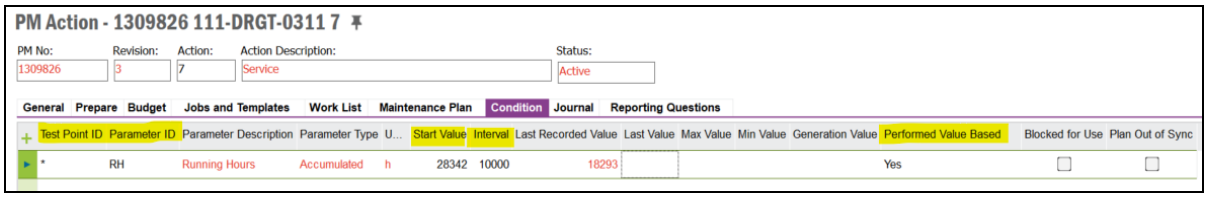



*Figure 14 – Running Hours Interval*

Regarding Lead Time, it must be filled in on the "General" tab and must follow the convention of this document for Lead Time:

| Final Interval Standard | Interval Unit | Leadtime in the IFS | Real Leadtime | Leadtime Unit |
|---|---|---|---|---|
| 2000 | Hours | 90 | 90 | Day |
| 4000 | Hours | 90 | 90 | Day |
| 8000 | Hours | 180 | 180 | Day |
| 16000 | Hours | 455 | 455 | Day |
| 24000 | Hours | 455 | 455 | Day |
| 25000 | Hours | 455 | 455 | Day |
| 50000 | Hours | 820 | 820 | Day |
| 60000 | Hours | 820 | 820 | Day |

*Table 2 – Running Hours Interval and Leadtime standards*

Lead time is a recommendation, and if the interval is not specified inside this table use the lead time of the most similar interval and conservative interval.

### 2.8.5 Maintenance Organization

The Maintenance Organization refers to the offshore department **A**ccountable for the work order preparation and closeout.

In the most of times this is also the **R**esponsible to perform the work as described through the Work Task Template (WTT). The maintenance organization will be defined at the Work Task Template level to guarantee standardization of disciplines for the fleet.

There is an exception for Mechanical which is defined generically for the WTT and will be defined between Mech Engine Room (MECHER) and Mechanical Topside (MECHTS) within the PM Action.

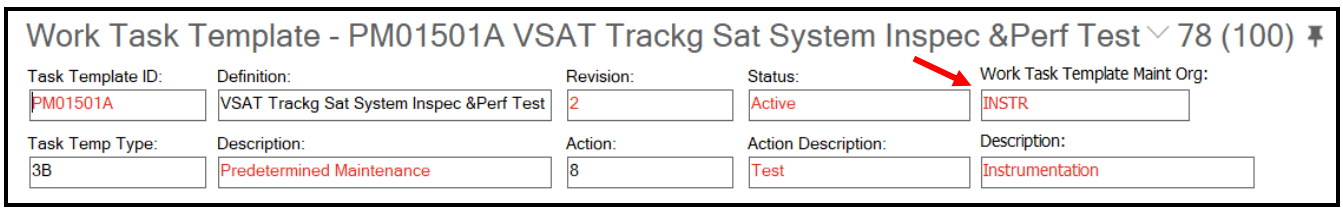



*Figure 15 – Work Task Template Maint Org*

The Work Task Template will be the guide for the PM Actions, where they must follow the defined main Organization recommendation, however there are exceptions, and this can be defined at the PM Action level (such as Mechanical Topside/Hull side and Metering for Brazilian Units)

Find below rules to set up the Maintenance Organization:

- LSA Code equipment should be under STCW (Cargo responsibility). Ex.: lifeboat, life raft, others.
- Function Test (that doesn't have maintenance task) should be performed by the operator, not maintenance. Ex.: Function Valve Test, PAGA function Test, Radio test, etc.
- For Radio Operator Function Tests as GMDSS, the W.O. will be derationed for CARGO as the STCW of the unit. The creation of Radio Operator can be considered.
- Intrusive maintenance (Inspection or Service) should oversee the department who will act (who have knowhow to perform).
- For departments who have same knowhow (mechanical T/S and E/R), the location should be taken in consideration (Ex. Valves, Pumps, etch).
- Equipment under NR13/SPIE, keeps with SPIE team.
- Equipment controlled by ANP, keeps with Metering.
- If doesn't fit above rules, it will be what is most practice in the most of units.

If the Job requires more than one discipline, the disciplines that are not Responsible for the Work Order are defined within the Resources tab, specified within the Work List of the Work Task Template.

### 2.8.6 Interval and Leadtime

The preventive maintenance interval is the specific period or amount of use between planned executions of preventive maintenance activities.

Leadtime is the definition of how long before the planned execution date the Work Order will be generated. Leadtime is a suggestion, the local responsible team can choose other values for a better planning and management.

It is important that the Work Order is generated before the due date, for planning resources, buying any required spare part, bring specifics tools offshore, etc.

The default Leadtime is defined based on the maintenance interval, as indicated in the table below. And as a default, the minimum value within the system is 90 days, so within the PM Action when 0 day is entered in the lead time, the real value is always +90 days.
Example: In IFS 90 days, the actual value is 90 days + standard 90 days = 180 days real lead time.

The interval is defined by the Equipment Maintenance Strategy (EMS) that follows the Interval Equivalence column, but for each different type of interval there are defined the following standards:

| Final Interval Standard | Unit Interval | EMS Interval Equivalence | Leadtime in the IFS | Real Leadtime | Leadtime Unit |
|---|---|---|---|---|---|
| 7 | Day | **1 Week** | 0 | 90 | Day |
| 8 | Day | **2 Week** | 0 | 90 | Day |
| 24 | Day | **1 Month** | 0 | 90 | Day |
| 54 | Day | **2 Month** | 0 | 90 | Day |
| 84 | Day | **3 Month** | 0 | 90 | Day |
| 174 | Day | **6 Month** | 0 | 90 | Day |
| 359 | Day | **12 Month** | 90 | 180 | Day |
| 724 | Day | **24 Month** | 365 | 455 | Day |
| 30 | Month | **30 Month** | 365 | 455 | Day |
| 36 | Month | **36 Month** | 365 | 455 | Day |
| 48 | Month | **48 Month** | 730 | 820 | Day |
| 60 | Month | **60 Month** | 730 | 820 | Day |
| 72 | Month | **72 Month** | 730 | 820 | Day |
| 84 | Month | **84 Month** | 730 | 820 | Day |

| | | | | | |
|---|---|---|---|---|---|
| 96 | Month | **96 Month** | 730 | 820 | Day |
| 120 | Month | **120 Month** | 730 | 820 | Day |
| 240 | Month | **240 Month** | 730 | 820 | Day |

*Table 3 – Interval and Leadtime Standards*

### 2.8.7 Action

The Action will be defined at the Work Task Template level, and the action will be passed to PM Action and becomes a directive of the Work Order. Below in the table are the Actions that must be chosen for preventive maintenance.

| Action Description | Action Code |
|---|---|
| Replace | 1 |
| Adjust | 4 |
| Service | 7 |
| Test | 8 |
| Inspection | 9 |
| Overhaul | 10 |
| Combination | 11 |

*Table 4 – Actions list.*

To consult the definition of each Action, follow "*ISO 14224 table B.5 — Maintenance activity*".
In relation to the Action "Combination" it must be used when we have a combination of activities as a Work Instruction, within SBM standards this is represented as being Items "5 - Detailed Inspection" and "6 - Function Test" of the Work Instruction.

### 2.8.8 Operational Status

The options used for the project will be defined between: Item Intrusive & Non-Intrusive.

- <u>Item Intrusive:</u>

Intrusive maintenance is maintenance that makes the function of the equipment or system unavailable.

Parts Replacement: Worn or damaged parts need to be replaced with new ones.
Instrument calibration: Instruments need to be calibrated to ensure measurement accuracy.

- <u>Non-Intrusive</u>

Non-intrusive maintenance is maintenance that does not make the function of the equipment or system unavailable.

Lubricating oil analysis: Analyze the lubricating oil to detect signs of wear or failure.
Thermography: Using a thermal camera to detect hot spots that may indicate failures.

Connecting "Operational Status" with "Action" field, there is the following relationship, where is used certain Operational Statuses for certain Actions.

| Action Description | Operational Status |
|---|---|
| Replace | Item Intrusive |
| Adjust | Item Intrusive or Non-Intrusive |
| Service | Item Intrusive or Non-Intrusive |
| Test | Item Intrusive or Non-Intrusive |
| Inspection | Item Intrusive or Non-Intrusive |
| Overhaul | Item Intrusive |
| Combination | Item Intrusive or Non-Intrusive |

*Table 5 – Relationship between Op Status and Action Description*

### 2.8.9 3rd party Allocation

For ease traceability it is usual to enable a check box on the Work Task Template and consequently on the PM Action and Work Orders which requires 3rd party outsourced resources or onshore support (e.g., external supply mobilization or repair & return routines).

3rd Party is defined as part of the service when:

- Strongly recommended as per SBM´s strategy.
- Required by Class, regulatory or certification.

The 3rd Party can also be defined at, by exception, PM Action level - Due to for example different regulations between regions. If necessary, this information can be differentiated at PM Action level.

Information is identified at Work Task Template level on the General tab or through reporting applications.

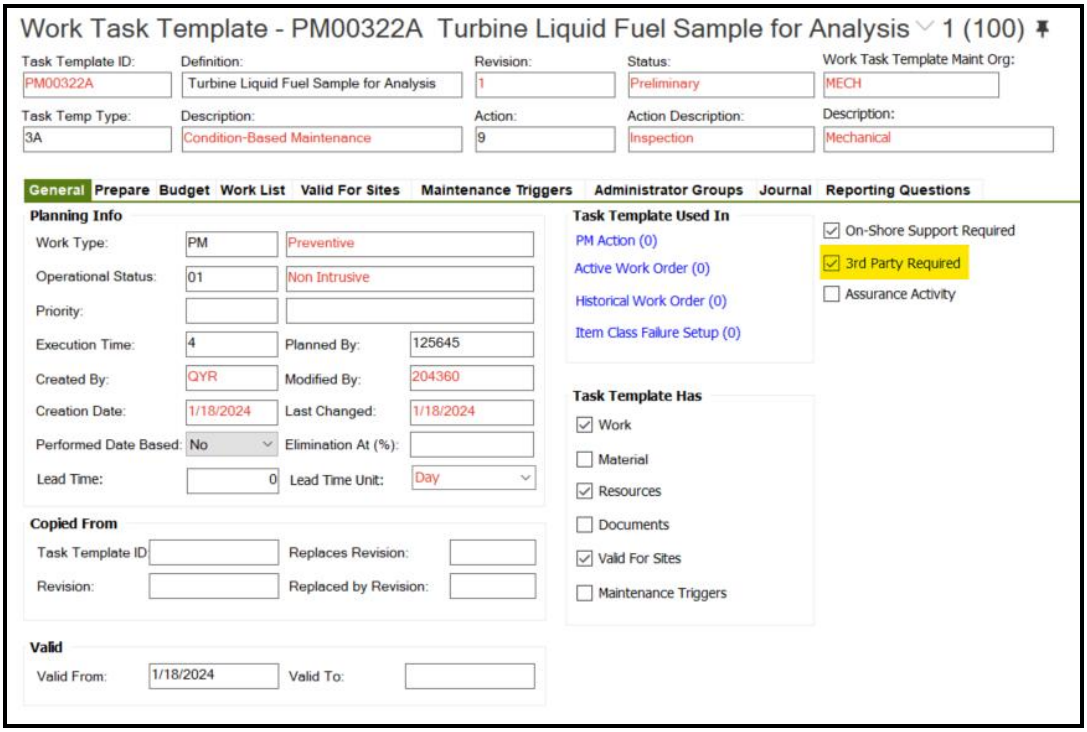



*Figure 16 - 3rd Party Required enabled checkbox.*

The way to check the 3rd Party for the Work Task Template is within the Work List tab, in the General tab it is read only information.

The "3rd Party Required" checkbox indicates that that service requires a third party in a very visual and easy-to-extract way. For more details about 3rd Party, check the Resource tab in the Work List.

When a 3rd party is flagged in WTT, it means there is a strong recommendation from the strategy team to use the 3rd party; it doesn't mean its use is mandatory – it can still be managed by the region.

When a third party is continuously involved due to contractual obligations and there is no need to involve new third parties, the Work Task Template recommends not using a third party.

### 2.8.10 Valid for Sites

The Valid for Sites is defined at the Work Task Template level, and this is where we define which ships that preventive maintenance task will be applied to.

If the ship is not specified within Valid for sites, it is not possible to create any PM Action for the objects on this ship.

Therefore, if the plan is applicable to the entire fleet, it must have all ship codes within its Valid for site.

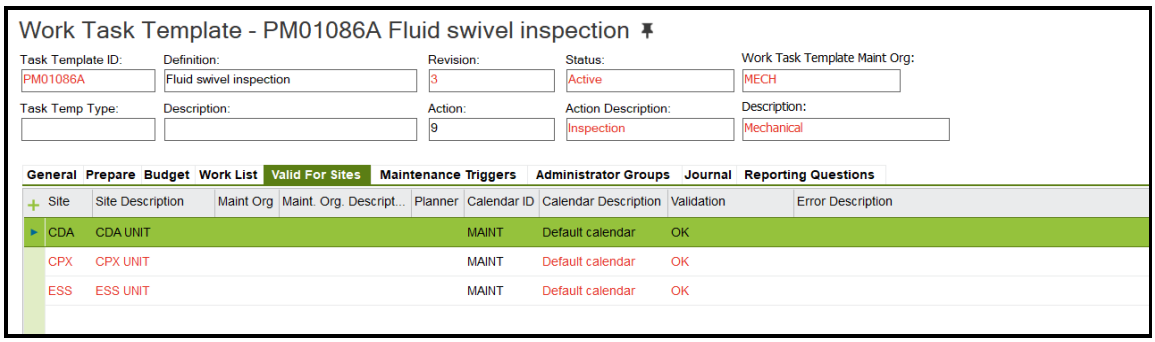



*Figure 17 – Valid for Sites (Only Turret's vessel)*

### 2.8.11 PM Group definition (Rounds)

The PM group is the tool used to group any number of PM actions into a work order as a route.

The PM group should be created when:

- The PM Group that doesn't last more than a workday and that aren't complex (Example: F&G detector test, damper test, etc.). There may be some exceptions for non-critical items such as manual valves, which are simple jobs that can take up to a week.
- The location of the objects should be considered (the Location Asset Tree could be used to guide the route).
- For components and instrumentation, equipment can be considered, e.g., instruments from the same equipment can be grouped together.

Note: Jobs that requires an individual report, parameter recording or certifications, should not be a round.

### 2.8.12 PM Group Standards

- PM Group ID:
  Starts with the ship's acronym, then a sequential number in order. Example: CDI00092
- PM Group Description:
  Briefly, what is the purpose of that route.

  As Standard for the Description: *Functional Description + Action + Location/Object Connected*

  Example: Fire Damper Function Test FZ02

  Note: Don't put specific parameters as Intervals in the Description.
- Site:
  Which ship does that route belong to. The routes are by ships and cannot be used on other ships.

At the PM Action level, it is defined whether it is part of a PM group or not, and to connect the PM Action to a PM Group, some rules are followed:

PMs need to have the attributes below:

- Same Interval.
- Same Maintenance Organization (Discipline).
- Same "Performed Date Based" parameter.
- Same date filled in "Due date" field for the next blank Maintenance Plan line.
- Same Priority.
- Same Lead Time.
- Same WTT - Work Task Template.

Note: Assurance activity SCE must not be mixed with BCE/PCE/NCE Type Equipment.

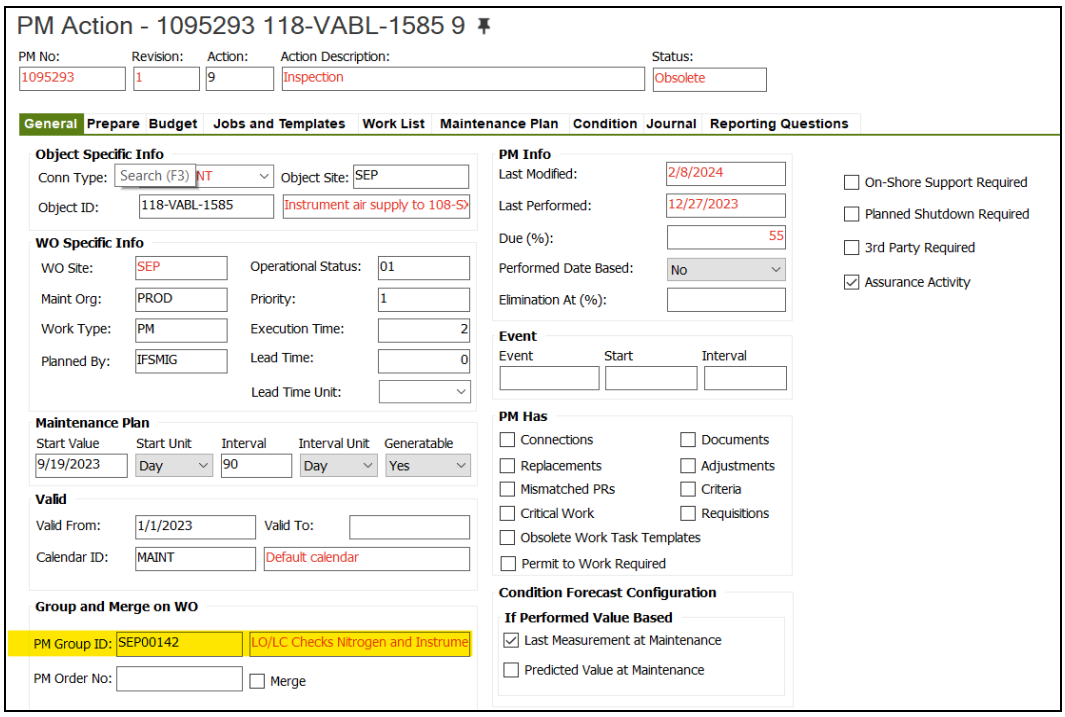



*Figure 18 – PM Group into PM Action*

### 2.8.13 PM Group Merge Standard

The PM Action has an option to “MERGE” the tasks, that directly interferes with the layout of the Work Order generated.

The option Merge must not be selected for any PM Action, the standard is:

- Not Merge for all the PM Actions.

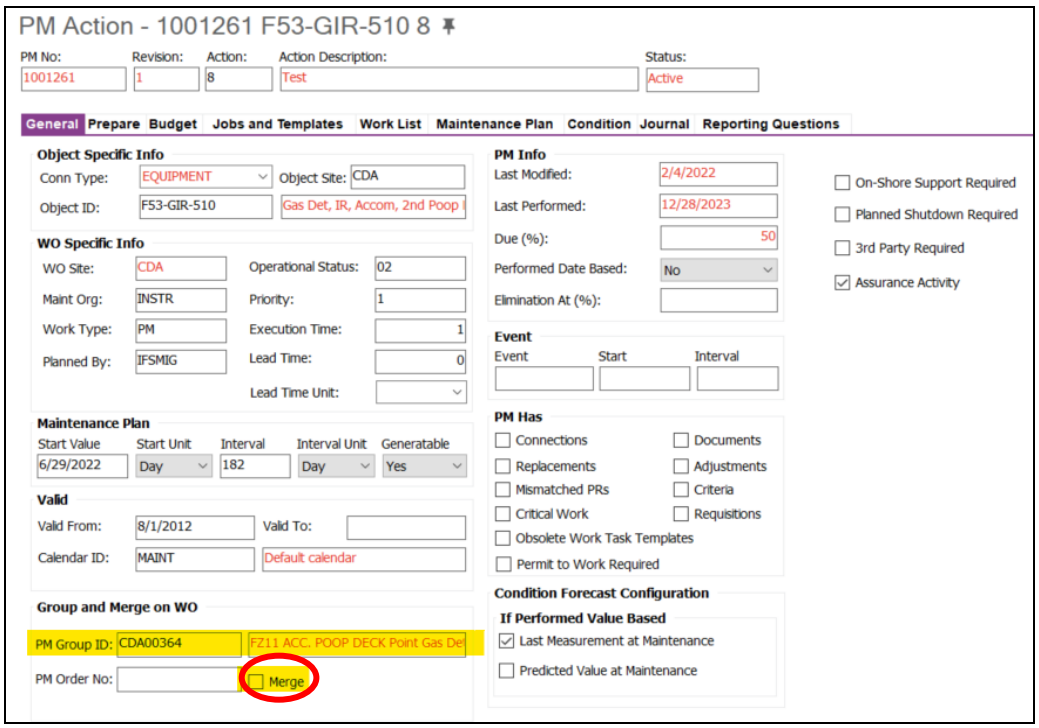



*Figure 19 – PM Action Not Merge*

Due to the default behavior of IFS regarding the Merge flag, when the user unchecks the Merge option in the PM Action, two situations can occur:

1. It is necessary to create a new PM Action if that PM has any open Work Orders.
2. If there are no open Work Orders, it is only necessary to review the Maintenance plan and ensure that the next scheduled maintenance is in accordance with the date of the last performed maintenance plus the required interval.

### 2.8.14 Shutdown Required

The "Shutdown Required" field is at PM Action level, and when selected it indicates that for that Job, on that specific Equipment, it requires the entire unit to be in Shutdown.

This field is important, mainly, for planning the activities that are carried out during the Unit Shutdown period and should be revised during W4W.

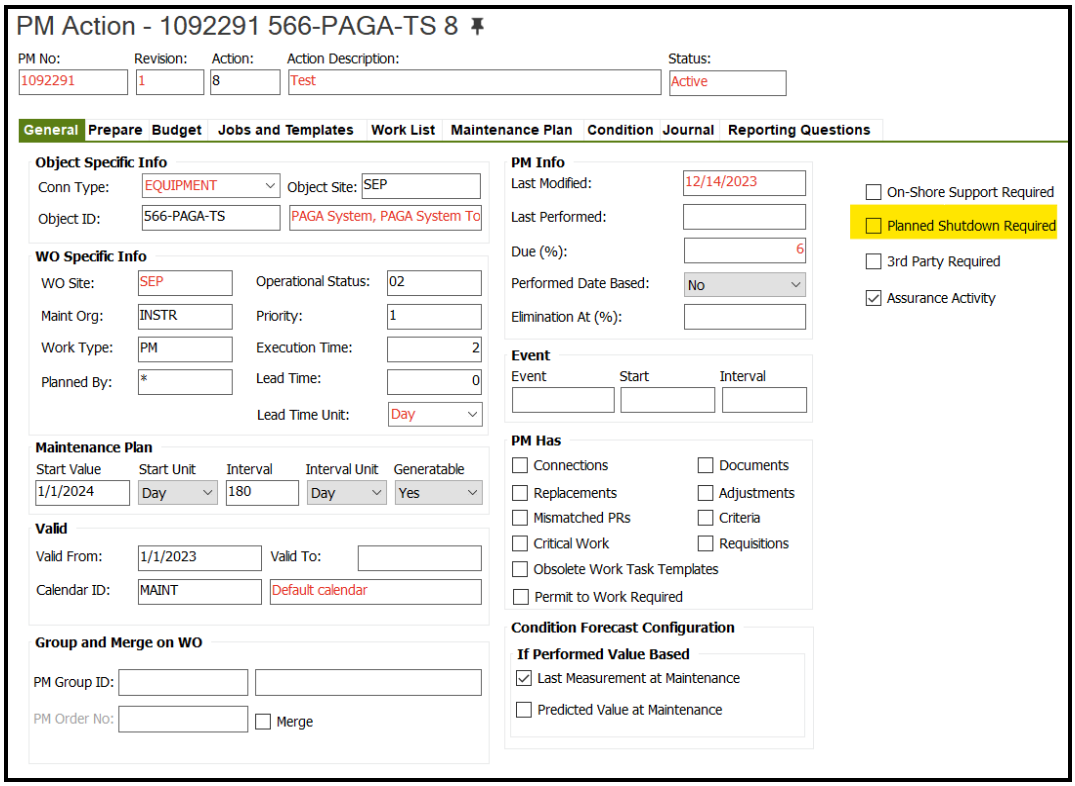



*Figure 20 – Shutdown Required Checkbox*

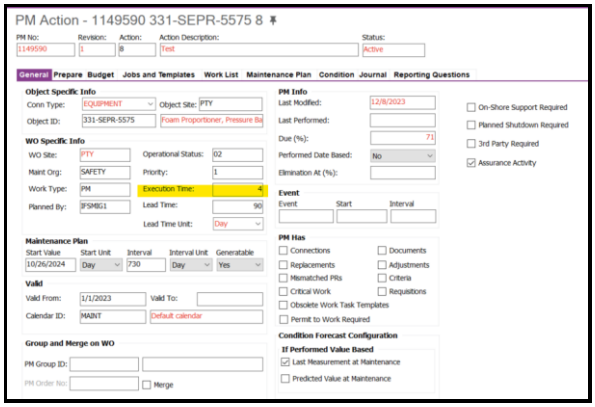
### 2.8.15 Execution Time and Duration

This field Execution time is used to specify the number of hours that you expect the work to take. Considering the in this period the work preparation (the isolation, the PTW issue, TRA…), and the after service (Paperwork, de-isolation, parameter recording, housekeeping…).

The Duration is the value of hands-on tool, that is, are considered only the hours spent on the maintenance. Excluding the pre and after work, like the Permission to Work and Paperwork after the service.

The Execution Time is in a PM Action level, and there are two rules to define it:

- If the equipment unavailability is less than 24 hours or the maintenance is non-Intrusive, the execution time and the duration must have the same value.
- If the equipment unavailability is more than 24 hours, the Work Task Template must be classified with an Enrichment Level 2 and the execution time, and the duration may have different value.



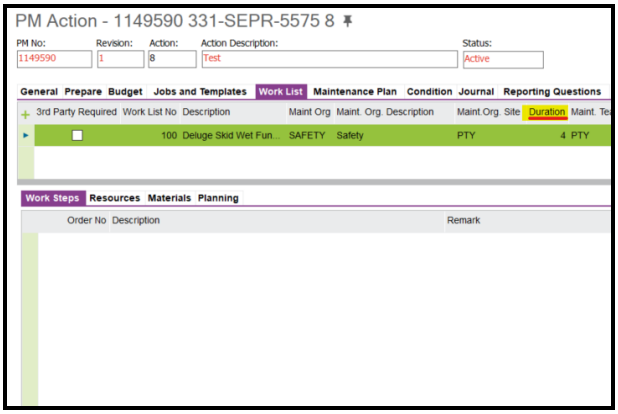




*Figure 21 & 22 – Execution Time & Duration in a PM Action level*

The Duration is defined in a Work Task Template level, inside the "Work List" tab, and the value entered in the Work Task template duration will be replicated to all the PM Action Work List duration.

The execution time in the PM Action General tab does not follow any parameter in the Work Task template, must be defined in the PM Action.

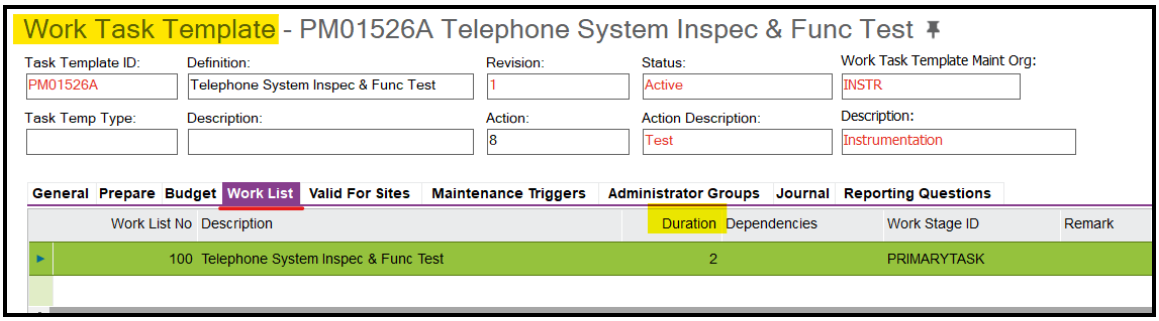



*Figure 23 – Duration defined in the Work Task Template*

### 2.8.16 Replacement Feature

IFS feature that allows a PM Action with a higher interval to replace the WO from another PM Action with a lower interval, in order to avoid overlapping PM Actions. Assuming that the scope of the PM with the higher interval already covers that of the lower one. With this, the system generates only one WO, reduces over maintenance with duplicate Work Orders and maintains more efficient planning.

Creates one WO from the higher interval PM Action for all PM Actions inside Replacement Structure.

For the less frequent PM Action to cover the more frequent PM Action, the Work Instructions must be complementary; that is, the less frequent PM must contain the scope of the other PM within its WI. If necessary, a Change Request to revise the strategy for this update should be submitted to the attached WI.

Once the "Replacement" functionality is selected for that maintenance strategy, it must be applied to all fleet-wide PM Actions.

Prerequisites for Preventive Maintenance to be classified as eligible for Replacement:

1. Performed Date Based "No"
2. PM Action status Active
3. Interval Unit different from "Days" – Use for Weekly, monthly and Year interval units. Attention in the Biweekly interval
4. Equal Lead Times
5. The least frequent PM should have the highest priority
6. Complementary Reporting Questions
7. Same Object ID
8. PM Actions with the Maintenance plans aligned

To build the Replacement structure, it is necessary to select the PM Action Active with the longest interval and select the "Replacement" option.

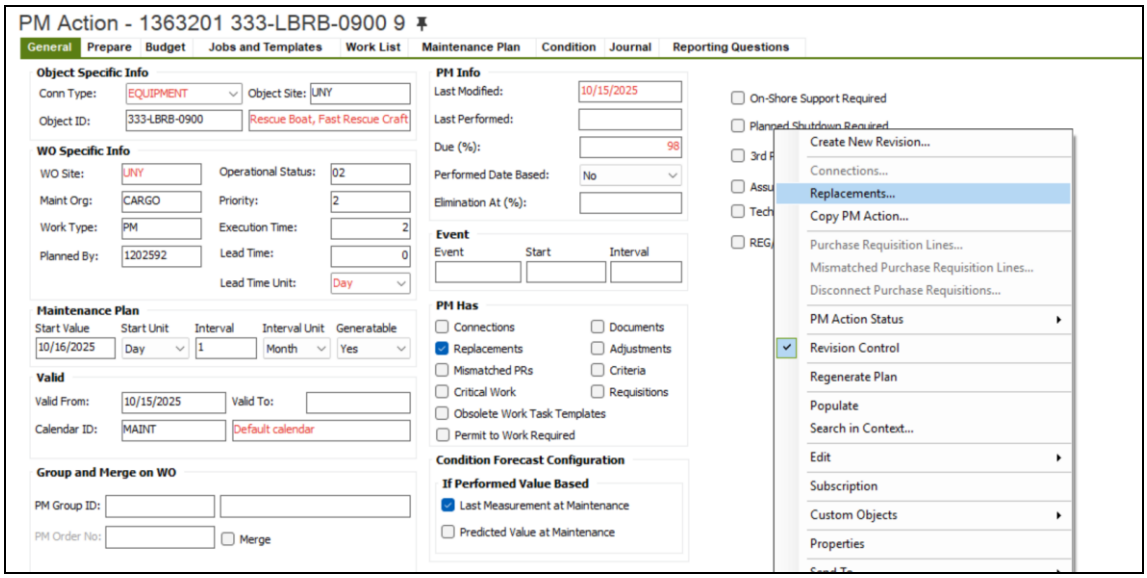



*Figure 24 – RMB for Replacement*

Next, we will add the structure to the PM Actions that will be replaced.

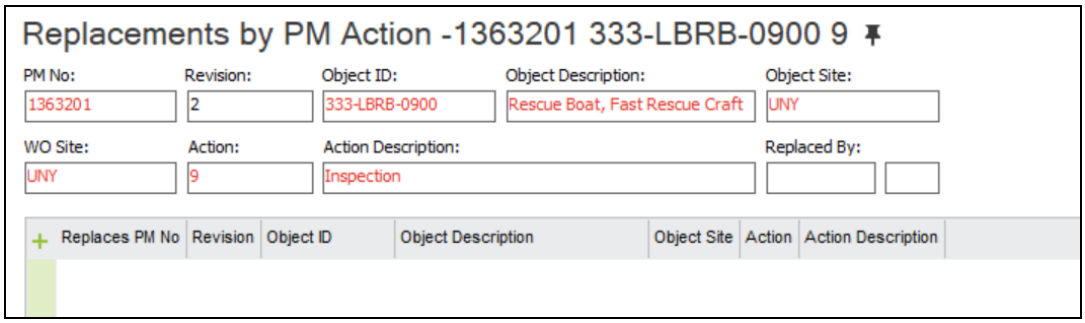



*Figure 25 – Replacement Structure*

After the structure is created, all PM Actions appear within the structure as follows:

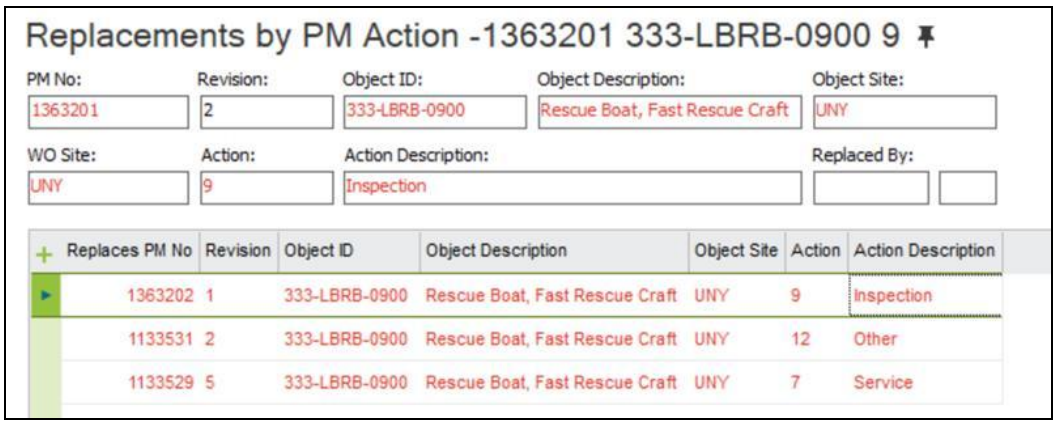



*Figure 26 – Replacement structure filled*

For each "Replacement" relationship, a specific structure must be built, for example:

Maintenance where there are 2 replacements: Semiannual / Monthly / Weekly

1. Must be created a structure where the Semiannual replaces the monthly and weekly.
2. The second mandatory replacement structure is for the monthly to replace the weekly.

### 2.8.17 Connections Feature

It is a planning feature that allows linking different Preventive Maintenance (PM) tasks, ensuring they are scheduled together. To achieve this, it is necessary to create a structure that synchronizes dependent scopes, such as shutdowns and overhauls. This approach is crucial for complex maintenance involving multiple teams, as it helps optimize resources and minimize downtime.

To create the structure, you need a Parent PM Action and Child PM Actions.

They should have:

1. The same interval
2. Performed date based “Yes”

To build the Connection structure, it is necessary to select the parent PM Action (Preliminary or Active – The Connection creates a new revision automatically) and select the "Connection" option. First select the superior PM Action (Parent), then connect the underlying PM Actions (Children) by using the right mouse button (RMB) in the PM Action then selecting the option “Connections”.

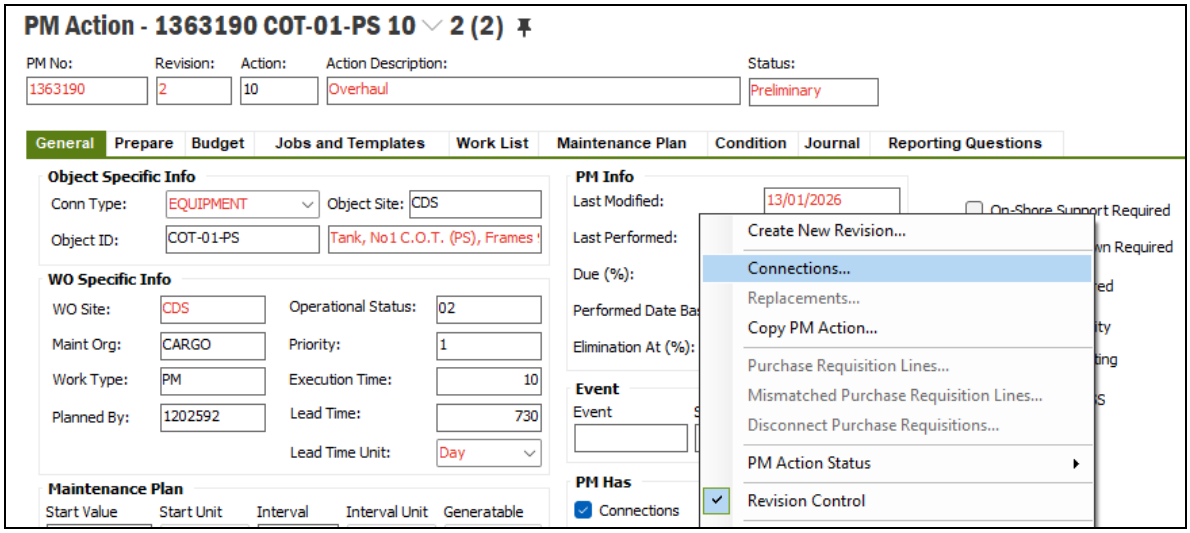



*Figure 27 – RMB for connections*

It will generate the PM Action Structure:

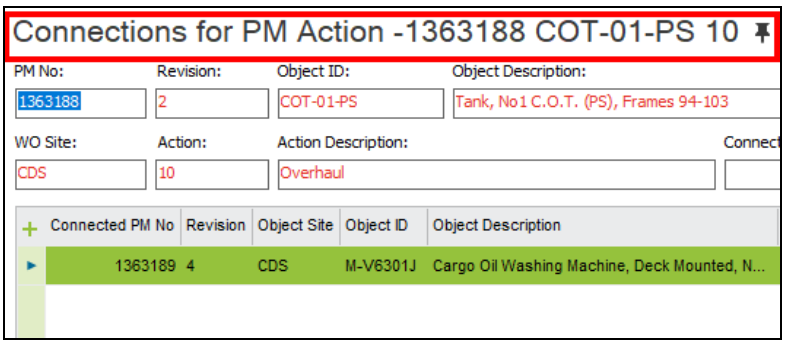



*Figure 28 – Connections Structure*

When the PM Action structure is built, those PMs WOs will be generated following the Connection functionality, represented by the WOs connection structure:

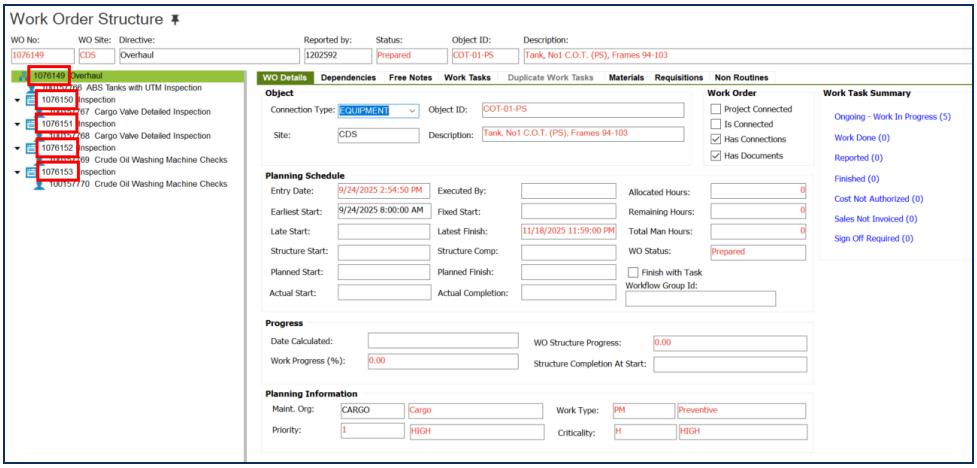



*Figure 29 – WO Connection struture*

The Parent PM Action will drive the Start Value, Interval, Interval Unit, Maint Plan dates and the Valid From of the Child PM Action and this reflect to the WOs in: Latest Finish, Earliest Start and Due dates. The Child Maint plan will always be a reflection of the Parent Maint plan.

Independent of the day you close the child WO, the next planned WO will follow the parent Planned dates.

## 2.9 PROJECT CHANGE REQUEST

### 2.9.1 Change Request General Attributes

Three types of Change Requests can be created during W4W:

- Strategy Review (when required).
- Structure review.
- PM Action review.

Below are the definitions of when to use each of the Change Request types.

- Change Request Site: 043C3 - Santos.
- Change Request Level: CRL3 – Essential
- Document Class: RP-026 – Change Request

### 2.9.2 Change Request Structure

If a change is required in the Structure, is required to open a Change Request in the CMMS to register the change. These changes will be considered:

- Creation of Functional Object or Serial Object
- Scrapped in Functional and Serial Objects (Only for Dummy Tags).
- Update to Item Class.
- Review of the FCA (Only if needed)

Important:

In the case of Scrapped in Objects that are not Dummy Tags, open a Change Request outside the project and ask that the first approval be from the onboard staff, confirming that the equipment is not in the field. Attach evidence, if possible, such as P&ID, layouts, etc.

For Functional Objects revision, it's not necessary to fill a reason, that's why is important to connect the Functional Object to the Change Request – That is the smartest way to have traceability.

It is not necessary to create Change Request for the cases below:

- Change of position of function object in the tree.

- Change the description of the object.
- Update the Object Structure tab ("Process" – Align with PSM Engineer).
- Update of Category field

*E3 "Discipline + Month" Structure Review*

For Change Requests of this type, the CR Type chosen must be CRT2 – New/Update Equipment

In the "Notes", there is a standard text:
"*Following the W4W project premises, see the attached files for more information about the changes made*."

Always attach the "As found" document from the engineer's analysis to the Change Request, to highlight the changes made. (Doc Class: RP-026)

When necessary to place the equipment in "Scrapped", connect the entire functional tree of the equipment that is changing status to the Change Request, this includes the children equipment. For Functional Objects, use the connection "Equipment Object" instead of any other "Equipment".

| Parameter | Object Connect |
|---|---|
| Functional Object | Equipment Object |

*Table 6 – Change Request*

**Important:**

1. The changes explained in the Change Request are only made when the Change Request has complete approval.
2. The first step must be approved by the Maintenance Strategy Discipline Lead and the second Step by the Maintenance Strategy Manager.
3. It's not necessary to have one Change Request per Item Class, the changes will be tracked by Batches per Discipline following the E3 Project calendar.
   Example: EI&C discipline on January will receive the Item Class to work against in that month – This is the Change Request content for EI&C in January.

### 2.9.3 Change Request PM Action Review

If a change is required in the PM Action, is required to open a Change Request in the CMMS to register the change. These changes will be considered:

- Used to update/create the PM Actions as required by engineer analysis in the project.
- Work Orders cancellation – Consider Priority 1, 2 and 3.

When reviewing the PM Action, fill in the "Reason" field with the CR number and Change Request Title.

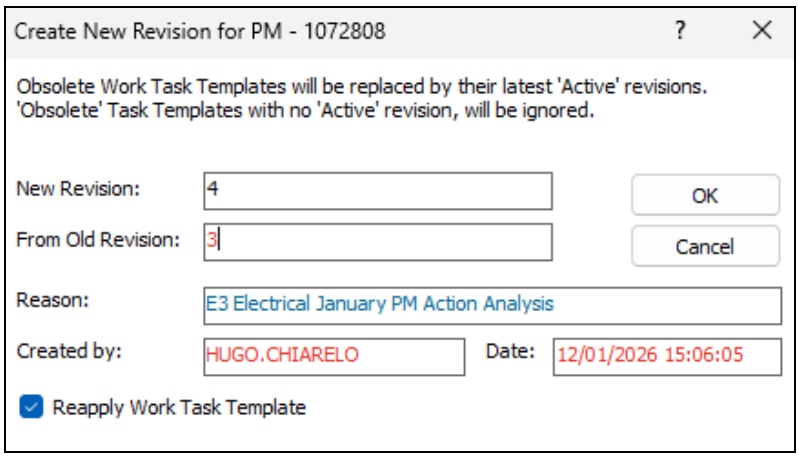



*Figure 30 – Revising the PM Action*

*E3 "Discipline + Month " PM Action Analysis*

For Change Requests of this type, the CR Type chosen must be CRT3 – New/Update PM Actions.

In the "Notes", there is a standard text:
"*Following the W4W project premises, see the attached files for more information about the changes made*."
If a Migration File is used during this process, record the Migration File ticket Number as standard text:
"See attached the Migration file for PM Actions update – Ticket No: *INCXXXXX*"

Always attach the "Reliability Release", "As found", "As left" and "Migration File" documents from the engineer's analysis to the Change Request, to highlight the changes made. (Doc Class: RP-026)
When necessary to connect Work Orders to Change Requests, use the "Active Separate" or "Historical Separate" category.

| Parameter | Object Connect |
|---|---|
| Active Work Order | Active Separate |
| Historical Work Order | Historical Separate |
| PM Action | PM Action |

*Table 7 – Change Request*

**Important:**

1. The changes explained in the Change Request are only made when the Change Request has complete approval.
2. The first step must be approved by the Maintenance Strategy Discipline Lead and the second Step by the Maintenance Strategy Manager.
3. For more than 200 PM Actions creation the Migration file tool is recommended.
4. During the review of PM Actions, do not reuse any PM Action for another WTT - If it is necessary to implement a new strategy on the equipment, do not use any existing PM Action, create a new one.
5. It's not necessary to have one Change Request per Item Class, the changes will be tracked by Batches per Discipline following the E3 Project calendar.
   Example: EI&C discipline on January will receive the Item Class to work against in that month – This is the Change Request content for EI&C on January.

### 2.9.4 Change Request Strategy Review

If after analyzing the EMS requires any change, open a Change Request in the CMMS to register the change. These changes will be considered:

- Create or Obsolete Work Task Templates.
- Create or update an Item Class in FASN – Boundaries, Maintenance Strategy Overview, Parent Item Class, Scope, Failure Modes, Causes and Maintainable Item.
- Revision in the Work Task template (Mandatory to fill the reason field with CR Number and motivation/what is the change).

When revising the Wok Task template, always put the CR Number and what is the change.

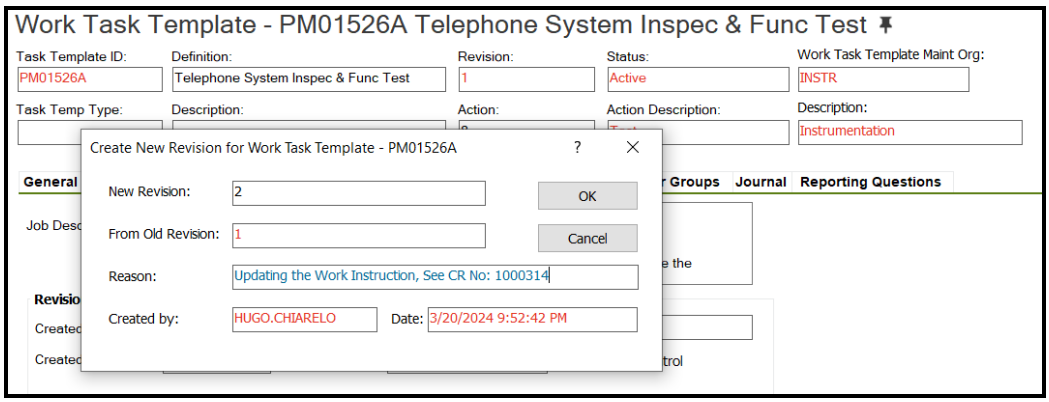



*Figure 31 – Work Task Template Reason field in the Revision*

It's not necessary to create Change Request for the cases below:

- Adequate some wrong information in the WTT and doesn't need to create a revision.

The standard Change Request Description for the Strategy Review:

*E3 "Item Class Code" Strategy Review*

For Change Requests of this type, the CR Type chosen must be CRT1 – New/Update EMS.

For changes that require editing the Performance Standard, must attach the approved SECE Governance form, to get traceability.

In the field "Notes" fill in which Work Task Templates or Item Classes are being reviewed, what the changes are, what are the motivations for the changes and what benefits these changes bring. See the questions that will guide how the Note should be specified in the CR.

A) What is the Change?
B) What is the motivation of the change?
C) Performance Standard Change or other Documentation change required?
D) What are the Benefits of change?
E) Does the change have impact on shutdown unit availability?
F) Is it feasible?
G) Is it going against previous change request?
H) Is it in compliance with local regulation or standards?
I) Actions Required

**Important:**

1. The changes just can be made with the CR fully approved.
2. The reliability engineer who initiated the request. The first approval must be done by the Reliability Lead must be the first approval, the Reliability Manager the second and the third approval must be the Reliability Lead responsible for the Item Class.
3. It's necessary to create a specific Change Request for each Item Class Maintenance Strategy change.
4. For Changes where you need to adequate wrong information in the WTT or FASN, but no change in the subject use the Change Request *XXXXXX - E3 Strategy review*

### 2.9.5 Change Order

The Change Order is created when the Change Request is fully approved, this indicates that the actions discussed in the CR can be executed and this execution is managed by the Change Order Module - CO.

In the CO it is necessary to put a summary of the actions that were carried out, insert this data in the "Work Done" field.

There is a standard text to be placed in Work Done for:

CR Analysis:

"*The PM Actions and Structure Update were done following the revision notes of the "As found" spreadsheet attached in the Change Request*".

CR Strategy Review:

"*The EMS was done following the Notes inside the CR and/or SECE Form attached for PS update*".

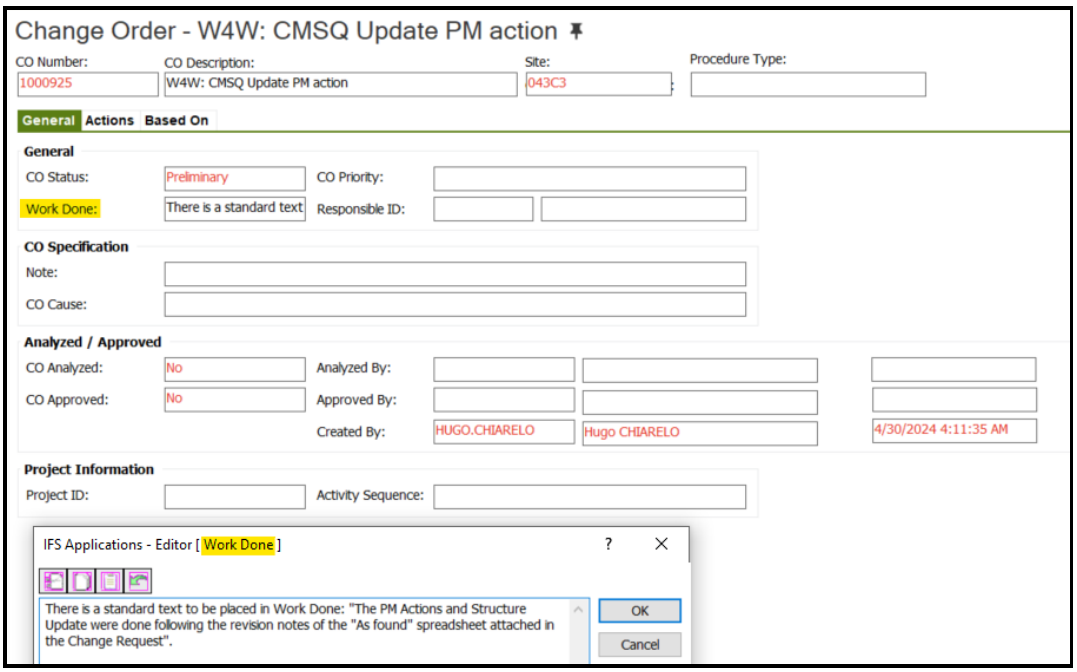



*Figure 32 – Work Done in the Change Order.*

It is also mandatory to fill out the person responsible for closing the Change Order - Reliability Engineer for CR Analysis and Reliability Senior Engineer for CR Strategy review.

If the execution of the task involves only one person in CO execution, it is not necessary to create the Change Order Line, the actions have already been specified in Work Done.

If the execution requires the participation of another person, a line must be created in the Change Order, and this person must be responsible for it. This line must describe the action that the person must perform. Possible actions are and their respective owners:

- Cancellation of Work Orders – Senior Reliability Engineer: *CR Analysis.*
- WOs parameters update (Latest Finish and Priority) – Process & System Reliability Engineer: *CR Analysis*.
- Performance Standard Update – PSM Reliability Engineer: *CR Strategy Review*.

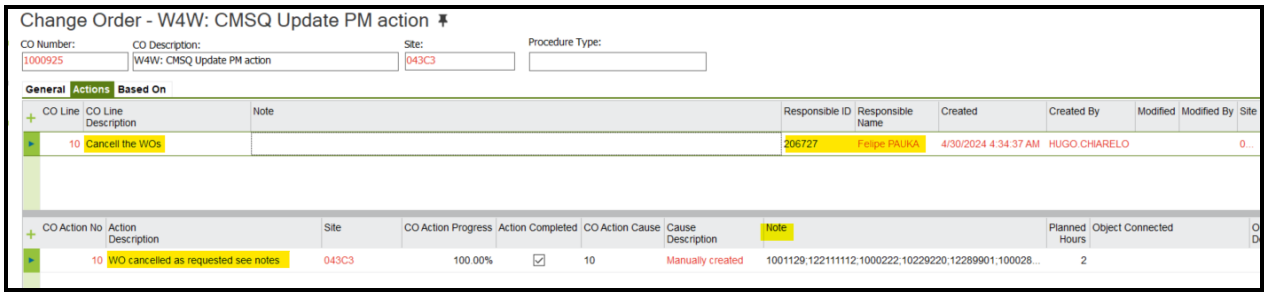



*Figure 33 – CO Lines and Actions Filled.*

More information about the actions executed can be found in the CO Action No "Notes".

After completing all actions, the Change Order must have its status changed to "Completed".

For Change Orders that require a review of Performance Standards, it is necessary to create a dedicated line item within the Change Order to control and monitor the actions related to Performance Standard

adjustments — including the local OTA responsible part and the duration of each action. This approach enables the establishment of KPIs and ensures systemic traceability of all changes

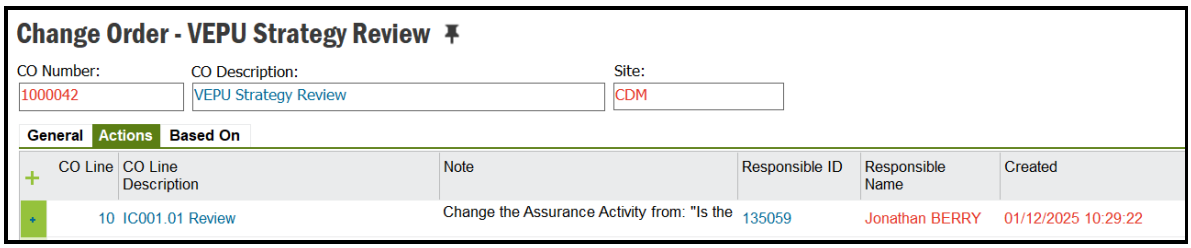



*Figure 34 – CO Lines and Actions with the OTA responsible.*

### 2.9.6 Cancelling Work Order

The Work Orders may be cancelled during reviewing PM Actions, but just the “Seniors” have the access to cancel W.O.

There are just three reasons to cancel a W.O.:

- During a change of Performance Date Based “NO” to “YES” – Use the option “Covered / Credited by another WO”. See the Chapter “2.8.2 and 2.8.3 Changing Performed date Based Standard” for more details how to manage this change.
- Because the Strategy is not applicable anymore (PM Action is fully obsolete) – Use the option “Cancelled following the Change Request”.
- If the Work Order has an incorrect Latest Finish.
  Note: For change manually the latest finish in the Work Order the acceptable range: up to 2 months difference with the correct date - Example: Preventive maintenance correct for some certification.
  Note 2: If there is a Historical Work Order and the correct Latest finish is bigger than the Maint plan calendar horizon, the manual Latest Finish is the right option.

For WOs Latest Finish that do not respect the range up to 2 Months, the following workflow adequate.

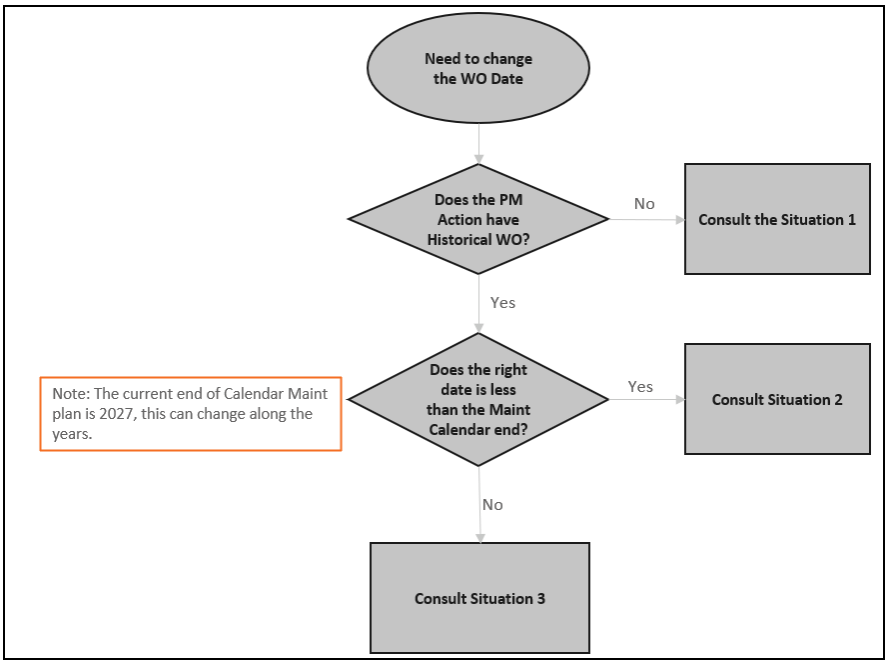



*Figure 35 – Cancel WO workflow.*

This is Situation 1:

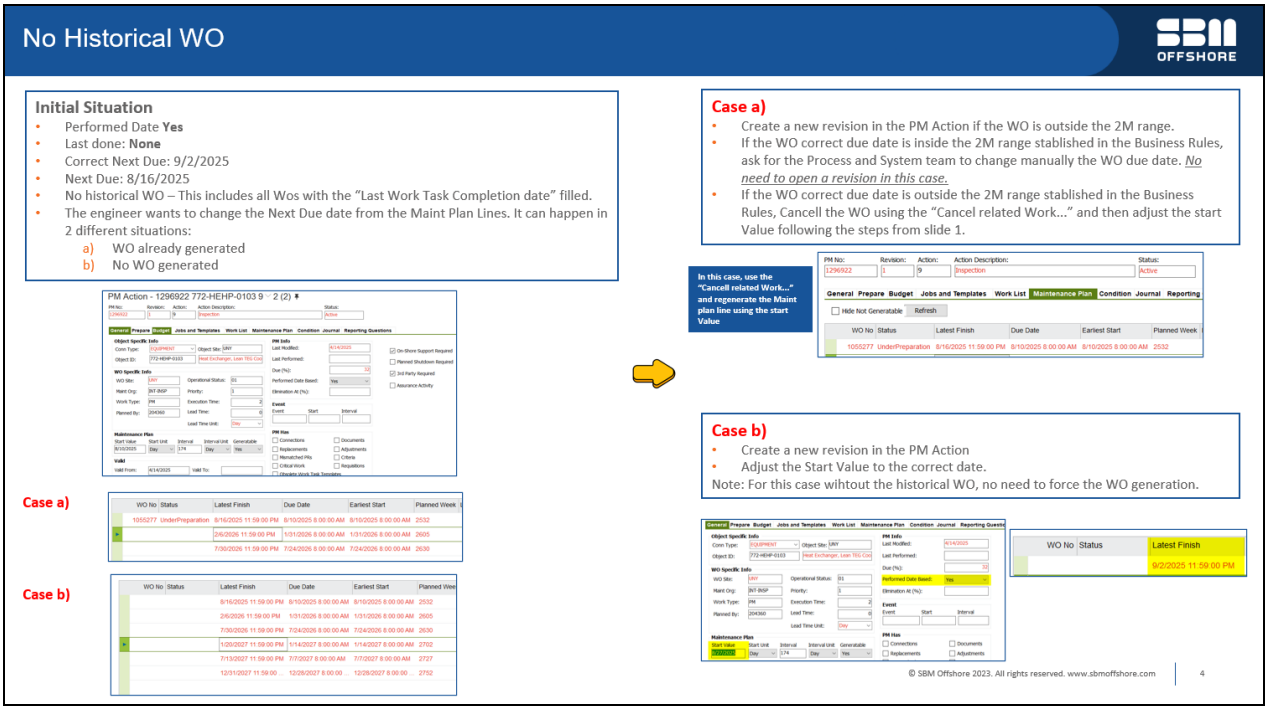



*Figure 36 – Cancel WO Situation 1*

This is situation 2:

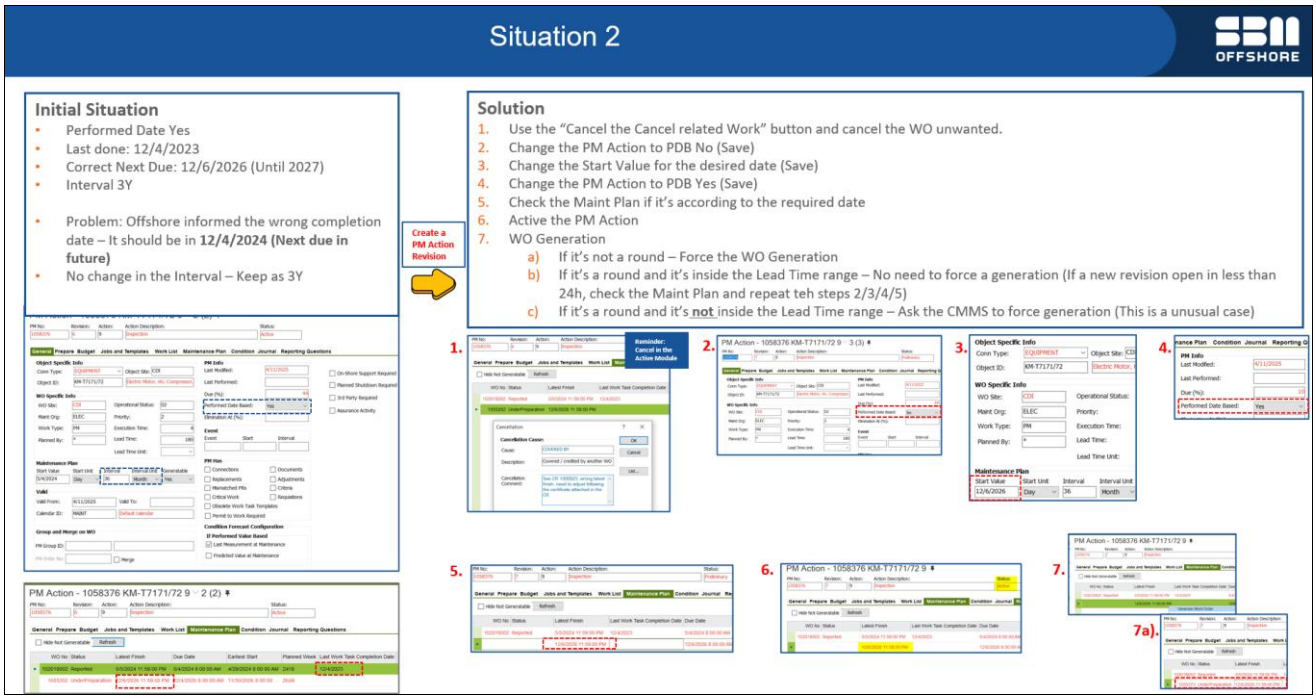



*Figure 37 – Cancel WO Situation 2*

This is the situation 3:

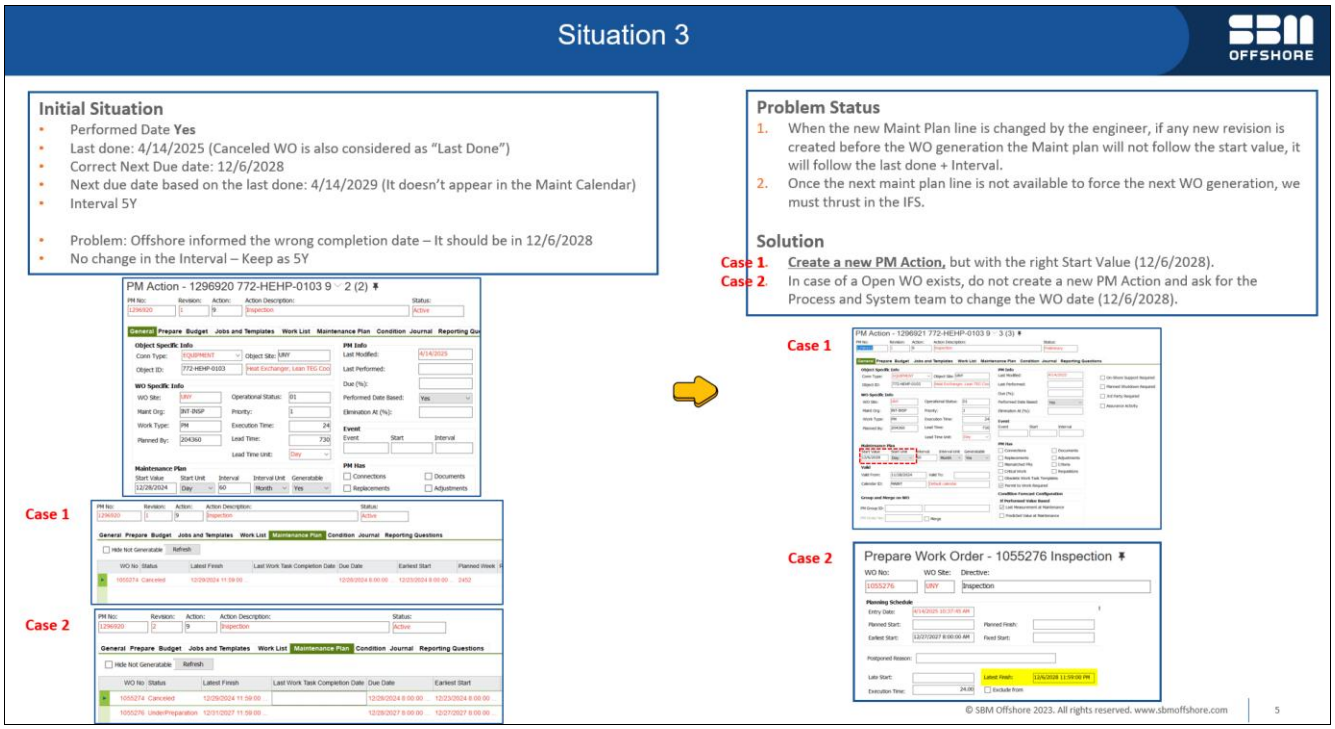



*Figure 38 – Cancel WO Situation 3*

Reasons to not cancel a WO for:

- Department (Maint Org) changes.
- Change Priority in the Work Order to comply with the PM Action.
- Latest finish on the Work Order - If the difference between the WO date and the correct date is less than or equal to two months difference or the correct Latest Finish is after the available Maint Plan horizon, there is no need to cancel. Latest finish is updated directly in the Work Order by Process and System reliability engineer.

For changes to Work Orders that do not require cancellation, the update actions performed will be recorded in the "Free Notes" tab.

The "Cancelation comment" field must be filled with the Change Request number and why the cancelation.

For Work Orders that must be canceled because they are covered by other Work Orders, the cancellation justification must contain the Work Order number that covers it.

Work Orders must be canceled on the "Active Work Orders" or "Prepare Work Order" screen, right click on the Header on the screen will display the options below:

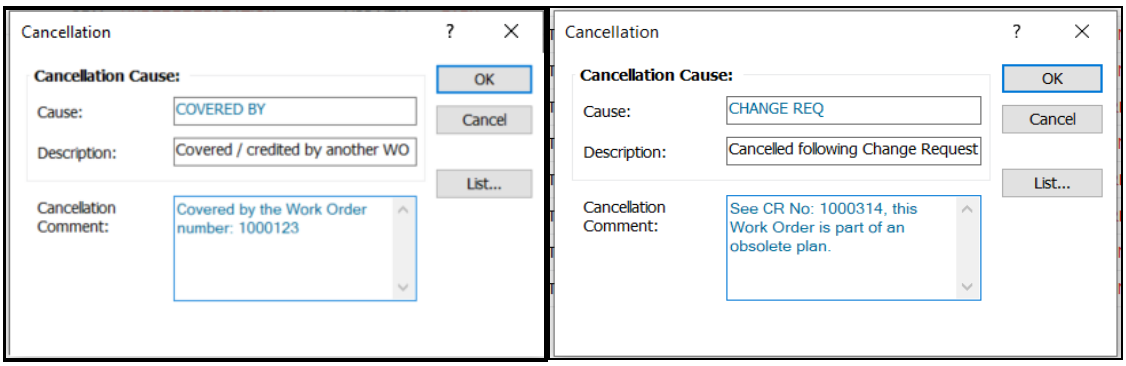



*Figure 23 – Canceling a Work Order in two different ways.*

In case of cancellation for preventive Work Orders, there is a screen to cancel through the Manit Plan using the "Cancel Related Work..." button.

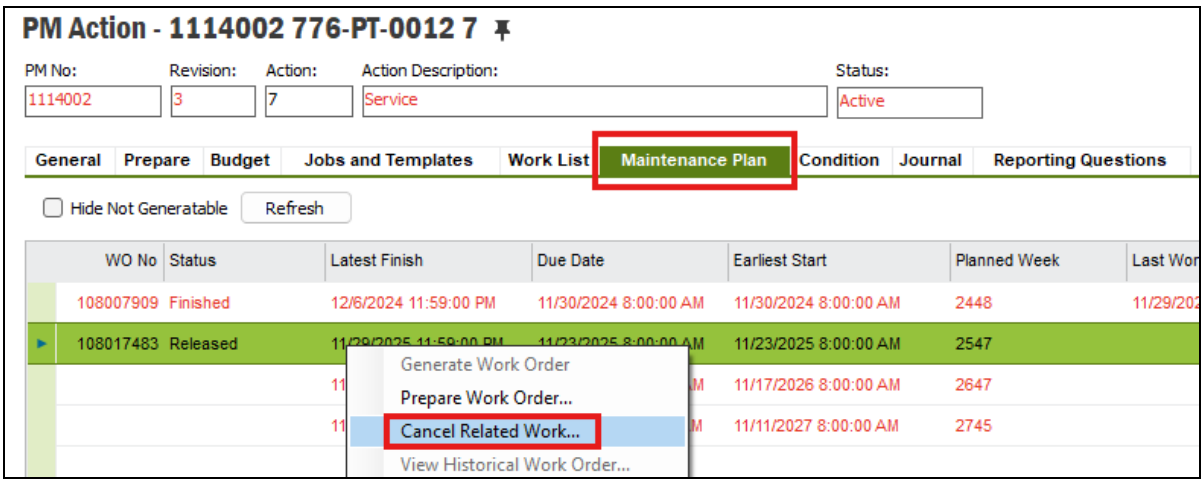



*Figure 24 – How to use "Cancel Related Work…"*

## 2.10 WHEN AND HOW TO CREATE NEW PM ACTION

Sometimes, you'll need to use the "Cancel Related Work…" function in the system. But if you're dealing with a big batch of PM Actions, there's an easier way — just create a new PM Action instead. This helps keep things organized when you have a lot to manage.

Here's what to do: set up your new PM Action and make sure its "Start Value" matches the old PM Action's maintenance plan. Then, for any old Work Orders you're replacing, you can cancel them right from the Active Work Order screen. You don't have to use "Cancel Related Work…" for each one.

Remember: whenever you cancel Work Orders with "Cancel Related Work…", you also need to cancel them in the Active Work Order module to keep everything up to date. This way, all your maintenance history stays with the Functional Object, not the PM Action itself.

The Equipment Maintenance history is kept in the Functional Object, not inside the PM Action.

Important: All the Wos canceled using the "Cancel Related Work…" needs to be canceled in the Active Work Order module too.

## 3. ANNEX AND REFERENCES


[1] ISO 14224, Petroleum, petrochemical and natural gas industries – Collection and exchange of reliability and maintenance data for equipment, 2016.

[2] CP-POM-MOC-OPS-PER-CMS.GRP.SWL.0000.ENG - Manage Changes in the CMMS

[3] Manage change / Preventive Maintenance Change Request (PMCR)

[4] OAPRMMWI999001 - FUNCTIONAL CRITICALITY ASSESSMENT